Optimal option market making and volatility arbitrage

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Abstract

We introduce a market making model of options which can encompass the trader's view on the underlying volatility versus the market implied volatility surface. An approximately optimal strategy is derived in closed form, where the optimal bid and ask levels depend on the expected volatility arbitrage profits associated with the quoted options. We show that the model can be extended to include additional features such as trading position limit, market making of arbitrary European derivatives, simultaneous market making of multiple options, and incorporation of risk control over customised factors.

Keywords: Market making, algorithmic trading, options, volatility arbitrage, Hamilton–Jacobi–Bellman equation.

1 Introduction

Options trading activity has experienced a phenomenal growth over the last few years. Bloomberg reported in October 2023 that the global volume of options trading across multiple asset classes has already surpassed that of futures.[1] The rapid increase of options turnover presents both a challenge and an opportunity to market makers, whose operation requires provision of competitive bid and ask quotes in split-second for a large number of option instruments (of different strikes and expiry dates on different underlyings) simultaneously. Efficient automation of option market making is therefore of great practical interests.

Algorithmic market making has received a lot of attention in the quantitative finance research community but existing works mostly focus on "delta-one" instruments such as single stocks. The standard modelling paradigm adopts a reduced form approach by assuming that the arrival rate of the market buy/sell orders can be influenced by the market maker via their posted level of spreads. The optimal market making strategy is then formulated as the solution to a stochastic dynamic programming problem. See, for example, Ho and Stoll (1981), Avellaneda

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[1]See "Options Are the Hottest Trade on Wall Street", Bloomberg, 12 October 2023. `https://www.bloomberg.com/news/articles/2023-10-12/options-are-the-hottest-trade-on-wall-street-after-crypto-demise`.

and Stoikov (2008), Guéant et al. (2013) and Cartea et al. (2014). Readers are also referred to the excellent textbooks of Cartea et al. (2015) and Guéant (2016).

Extension of the existing approaches to option market making, however, is far from being trivial. The non-linearity of the option instrument creates exposure to additional risk factors. Even under perfect delta-hedging, the market maker's profit-and-loss (PnL) will still be exposed to gamma-theta tradeoff and implied volatility movement. The resulting optimisation problem thus has a rather different structure, where the (actual and implied) volatility of the underlying will play an important role. Studies dedicated to option market making are relatively scarce. In Stoikov and Sağlam (2009), the market maker is assumed to trade in a single European call option under different modelling setups and the optimal quotes are obtained by discrete time approximation. El Aoud and Abergel (2015) look at optimal option market making of a single option under a stochastic volatility model. Baldacci et al. (2021) also consider a stochastic volatility framework but they further focus on market making of multiple options, and it is shown that dimension reduction is possible upon an approximation that the vega of each option is constant over the (short) market making duration. On the empirical front, Lai et al. (2023) show that Hawkes process can model option market order flow satisfactorily and they approximate the optimal quotes by a deep neural network.

In this paper, we present an option market making model based on Avellaneda and Stoikov (2008) where the arrival of option market orders is modelled by a Cox process which intensity depends on the posted spreads by the market maker. Unlike El Aoud and Abergel (2015) and Baldacci et al. (2021) who specialise to a stochastic volatility model, we take a different modelling approach by allowing the whole market implied volatility surface to be an input to our algorithm. Specifically, our framework facilitates expression of the market maker's view on volatility via the difference between their estimated (short term) actual volatility relative to the market implied volatility, which is natural from the perspectives of derivatives trading and volatility arbitrage.[2] In a baseline setup featuring a single option only, we derive closed form expressions for the optimal bid and ask quote under a second-order approximation. The optimal quotes admit a financially intuitive decomposition as a sum of three components: 1) the expected (and "penalty-discounted") volatility arbitrage profit that the market maker can harvest based on the actual-versus-implied volatility differential; 2) a quantity linked to the demand/supply elasticity governing how sensitive the order flow is with respect to change in the quoted spreads; and 3) a risk-adjustment term related to the trader's current inventory and the penalty level imposed on the running and residual inventory.

We then present a number of extensions under which the optimal spreads are still analytically tractable, sometimes under additional approximations such as the dollar gamma of the option remains unchanged over the market making horizon. Features that one can incorporate include hard-limit on the option position, market

[2]Existing models based on stochastic volatility, such as El Aoud and Abergel (2015) and Baldacci et al. (2021), imply that the market maker can only express their view via different estimators of the drift and volatility of the stochastic variance. Implicitly, it is assumed that the market maker and the market agree on the spot volatility of the underlying. Hence there is no gamma PnL within the market maker's wealth dynamics in such models and only vega PnL is present. In contrast, gamma PnL has a first-order importance in our framework, and we will show that our optimal bid/ask quotes are directly related to the volatility arbitrage profit which scales with the option gamma.

making of general European derivatives, market making of multiple options, and a versatile factor-based approach to control the option portfolio risk. Finally, we demonstrate the practical implementation of the option market making algorithm via several numerical examples.[3]

2 The baseline model

There is a market maker who provides pricing quotes on a European call option with a given strike $K > 0$ and maturity $\tau > 0$ written on a risky asset. Denote by $S = (S_t)_{t\geq 0}$ the price process of the underlying risky asset. From the perspective of the market maker, S has dynamics of

$$\frac{dS_t}{S_t} = \sigma_t dB_t,$$

where B is a standard one-dimensional Brownian motion. The market maker does not have any view on the (short-term) asset drift while $\sigma = (\sigma_t)_{t\geq 0}$ is his subjective assessment of the asset volatility process. In practice, $(\sigma_t)_{t\geq 0}$ can originate from some volatility estimation/prediction procedure beyond the specification of our market making model. We assume $\sigma_t = \sigma(t, S_t)$ for some function $\sigma(\cdot, \cdot)$ such that it has a local volatility form (and we will see a further specialisation of constant volatility can lead to closed form solution). Dividend and interest rate are assumed to be zero throughout this paper.[4]

Option positions are evaluated on a mark-to-market basis according to a given implied volatility surface $\sigma_{imp} = \sigma_{imp}(K, \tau)$. For practical implementation, σ_{imp} can either be inferred from the market (mid) prices of traded options or a suitably calibrated parametric model of the implied volatility surface. We assume σ_{imp} to be unchanged over the market making horizon. This assumption is normally acceptable as the implied volatility surface is typically recalibrated only daily or at most few times a day while the market making horizon can be much shorter (e.g. up to several hours only).[5] By definition of implied volatility, the mark-to-market price of the call option at time t is given by

$$O_t = O(t, S_t) := BS_{call}(S_t, K, \tau - t, \sigma_{imp}(K, \tau)),$$

where BS_{call} is the Black-Scholes formula for call option. Alternatively, the function $O(t, s)$ can also be characterised by the solution to the Black-Scholes PDE

$$\partial_t O + \frac{\sigma_{imp}^2}{2} s^2 \partial_{ss} O = 0, \qquad (t, s) \in [0, \tau) \times (0, \infty), \tag{1}$$

subject to the terminal condition $O(\tau, s) = (s - K)^+$.

[3]The relevant codes can be found in the following Google Colaboratory notebook: `https://colab.research.google.com/drive/1SxWvqWxwiYDdupN2ReuVI6HA52Arkmpn?usp=sharing`.

[4]This assumption can be relaxed to that short-term dividend and interest rate (up to the market making horizon) are zero.

[5]Vanilla options traders in practice seldom operate with a fully dynamic implied volatility surface (e.g. via a parametric stochastic volatility pricing model) since mark-to-market of positions can be more efficiently achieved by a well-calibrated static implied volatility surface. The risks associated with implied volatility surface movement are typically monitored via vega bucket sensitivity rather than being internalised by some stochastic volatility model. We will show in Section 4.3 how exposure to stylised implied volatility surface movement can be minimised.

Market making takes place over a time interval $[0, T]$ with $T \in (0, \tau)$. At each time t, the market maker simultaneously posts limit buy and sell order for one lot of the option at spread level δ_t^b and δ_t^a respectively around the mark-to-market option price, i.e. the bid and ask price are $O_t - \delta_t^b$ and $O_t + \delta_t^a$ respectively. Without loss of generality, the lot size is assumed to be one. Let Y_t^b and Y_t^a represent the cumulative number of options that have been bought and sold respectively in the time interval $[0, t]$, and write Q_t as the market maker's net option inventory at time t where we assume $Q_0 = 0$. Then by definition we have $Q_t = Y_t^b - Y_t^a$.

We model $Y^b = (Y_t^b)_{t\geq 0}$ and $Y^a = (Y_t^a)_{t\geq 0}$ as Cox processes with state-dependent intensities $\lambda_t^b = \lambda^b(\delta_t^b)$ and $\lambda_t^a = \lambda^a(\delta_t^a)$. The intensity functions are chosen such that $\lambda^i(\cdot)$ is decreasing for $i \in \{b, a\}$, and in turn the process Y^i is more likely to jump at time t when δ_t^i is small. Economically, it captures the behaviour that a limit order is more likely to be filled when the quote is competitive. Following Avellaneda and Stoikov (2008), the intensity functions are assumed to have an exponential form $\lambda^i(\delta^i) = \lambda_0^i e^{-\kappa^i \delta^i}$, where $\lambda_0^i, \kappa^i > 0$ are some constants for $i \in \{b, a\}$. λ_0^i reflects the average number of arriving market orders executed at the "fair" mark-to-market price per unit time, while κ^i captures the elasticity of demand and supply such that the average number of arriving orders per unit time will drop by a fraction of $e^{-\kappa^i}$ when the quoted spread increases by one unit from zero.[6]

The market maker is required to delta-hedge as part of their operational mandate, such that at time t they short Δ_t units of the underlying asset with $\Delta_t := Q_t \partial_s O(t, S_t)$. The quantity $\partial_s O(t, S_t)$ is simply the Black-Scholes delta of the call option. Importantly, the computed delta is based on the implied volatility σ_{imp}, rather than realised volatility[7] – the way to choose the "best" delta for the hedging purposes has been a long-standing research problem (e.g. Gatheral (1997)), and in practice often incorporates adjustments based on proprietary market maker's algorithms.

The cash process of the market maker, denoted by $X = (X_t)_{t\geq 0}$, follows the dynamics

$$X_t = \int_0^t (O_u + \delta_u^a) dY_u^a - \int_0^t (O_u - \delta_u^b) dY_u^b + \left[S_t \Delta_t - S_0 \Delta_0 - \int_0^t \Delta_u dS_u \right].$$

The first term represents the total sale proceeds when the ask prices are lifted, the second term is the total purchase costs when the bid prices are hit, and the last term is the cash flow from adjusting the delta-hedge. The market maker's portfolio consists of cash, options and stocks, which mark-to-market value at time t is given by $V_t := X_t + Q_t O_t - \Delta_t S_t$. Its process can be written as

$$\begin{aligned}
dV_t &= d(X_t + Q_t O_t - \Delta_t S_t) \\
&= dX_t + Q_t dO_t + O_t dQ_t - \Delta_t dS_t - S_t d\Delta_t - dS_t d\Delta_t \\
&= (O_t + \delta_t^a) dY_t^a - (O_t - \delta_t^b) dY_t^b + S_t d\Delta_t + d\Delta_t dS_t + Q_t \left[\partial_t O(t, S_t) dt + \partial_s O(t, S_t) dS_t + \frac{1}{2} \partial_{ss} O(t, S_t)(dS_t)^2 \right] \\
&\qquad + O_t (dY_t^b - dY_t^a) - Q_t \partial_s O(t, S_t) dS_t - S_t d\Delta_t - dS_t d\Delta_t \\
&= \delta_t^a dY_t^a + \delta_t^b dY_t^b + Q_t \partial_t O(t, S_t) dt + Q_t \frac{\sigma_t^2}{2} S_t^2 \partial_{ss} O(t, S_t) dt \\
&= \frac{\sigma_t^2 - \sigma_{imp}^2}{2} Q_t S_t^2 \partial_{ss} O(t, S_t) dt + \delta_t^a dY_t^a + \delta_t^b dY_t^b,
\end{aligned}$$

[6] For discussion on the estimations of λ_0^i and κ^i from market data, see for example Laruelle (2013) and Fernandez-Tapia (2015).

[7] Delta-hedging options using delta computed with realised and implied volatility will typically result in very different PnL profiles. See Ahmad and Wilmott (2005).

where we have used the construction that $\Delta_t := Q_t \partial_s O(t, S_t)$, and the Black-Scholes PDE (1) which allows us to express the option theta $\partial_t O$ in terms of its gamma $\partial_{ss} O$. The interpretation of the portfolio gain dV_t is financially natural: the first term is the volatility arbitrage profit depending on the realised-implied variance spread and the option gamma (also known as the "gamma-theta carry"), while the second and third term are the spreads earned by the market maker when a bid/ask quote is filled.

The market maker's performance criterion is

$$\mathbb{E}\left[V_T - \beta \int_0^T Q_u^2 du - \alpha Q_T^2\right], \tag{2}$$

which is the expected value of the terminal portfolio value penalised by the terminal and running warehouse inventory for some penalty parameters $\alpha \geq 0$ and $\beta \geq 0$. Here, α controls the risk level at the end of a trading episode while β controls the intraday risk exposure throughout the entire market making duration (and is linked to risk aversion level of an exponential utility function). Their values can be fine tuned according to the underlying risk management policy. In practice, one may wish to penalise the greeks of the option positions instead. To discourage vega exposure, for example, one can consider a penalty term in form of $\tilde{\alpha}(Q_T \text{Vega}_T)^2$, where Vega_T is the terminal vega of the option. With a short market making horizon T we may assume that the terminal vega is approximately the same as the initial vega Vega_0 (this kind of approximation is considered by Baldacci et al. (2021) as well). Then this can be incorporated upon replacing the α in (2) by $\tilde{\alpha}\text{Vega}_0^2$. The same can be done for β. More generally, the penalty parameters α and β in (2) can be chosen to scale with the option greeks.

The objective of the market maker is to maximise (2) with respect to the spreads posted $\delta^a = (\delta^a_t)_{t\geq 0}$ and $\delta^b = (\delta^b_t)_{t\geq 0}$. Mathematically, they seek a solution to the dynamic optimisation problem

$$h(t,s,q) := \sup_{\delta^b,\delta^a} \mathbb{E}\left[\int_t^T \left(\frac{\sigma_u^2 - \sigma_{imp}^2}{2} Q_u S_u^2 \partial_{ss} O(u, S_u) du - \beta Q_u^2 du + \delta_u^a dY_u^a + \delta_u^b dY_u^b\right) - \alpha Q_T^2 \middle| S_t = s, Q_{t-} = q\right], \tag{3}$$

where the admissible controls $\delta^i = (\delta^i_t)_{t\geq 0}$ for $i \in \{b, a\}$ are some predictable and integrable processes such that $\mathbb{E}[\int_0^T |\delta^i_u| du] < \infty$.

3 Solution to the option market making problem

Instead of using native gamma, it is more convenient to express the option's second-order risk via its *dollar gamma* (evaluated at time t when the spot is s) which is defined as

$$\Gamma^\$(t,s) := s^2 \partial_{ss} O(t,s) = \frac{K\phi(d_2(t,s))}{\sigma_{imp}\sqrt{\tau - t}},$$

where $\phi(\cdot)$ is the probability density function of a standard normal random variable and

$$d_2(t,s) := \frac{\ln \frac{s}{K} - \frac{\sigma_{imp}^2}{2}(\tau - t)}{\sigma_{imp}\sqrt{\tau - t}}.$$

Economically, dollar gamma reflects the change in the delta notional per unit change in the underlying price. Problem (3) can then be rewritten as

$$h(t,s,q) := \sup_{\delta^b,\delta^a} \mathbb{E}\left[\int_t^T \left(\frac{\sigma_u^2 - \sigma_{imp}^2}{2} Q_u \Gamma^{\$}(u,S_u)du - \beta Q_u^2 du + \delta_u^a dY_u^a + \delta_u^b dY_u^b\right) - \alpha Q_T^2 \middle| S_t = s, Q_{t-} = q\right].$$

The associated Hamilton-Jacobi-Bellman (HJB) equation is

$$\begin{aligned}\partial_t h + \frac{\sigma^2(t,s) - \sigma_{imp}^2}{2}\Gamma^{\$}(t,s)q - \beta q^2 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}h + \sup_{\delta^a}\{\lambda^a(\delta^a)[\delta^a + h(t,s,q-1) - h(t,s,q)]\} \\ + \sup_{\delta^b}\left\{\lambda^b(\delta^b)[\delta^b + h(t,s,q+1) - h(t,s,q)]\right\} = 0, \end{aligned} \tag{4}$$

with terminal condition $h(T,s,q) = -\alpha q^2$. Under our specification of the intensity functions $\lambda^i(\delta^i) = \lambda_0^i e^{-\kappa^i \delta^i}$ for $i \in \{b,a\}$, the optimal form of δ^b and δ^a can be pinned down from the first order condition such that

$$\delta^{a,*} = \frac{1}{\kappa^a} + h(t,s,q) - h(t,s,q-1), \qquad \delta^{b,*} = \frac{1}{\kappa^b} + h(t,s,q) - h(t,s,q+1), \tag{5}$$

where then the HJB equation (4) now becomes

$$\begin{aligned}\partial_t h + \frac{\sigma^2(t,s) - \sigma_{imp}^2}{2}\Gamma^{\$}(t,s)q - \beta q^2 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}h + \frac{\lambda_0^a e^{-1}}{\kappa^a}\exp\left[-\kappa^a(h(t,s,q) - h(t,s,q-1))\right] \\ + \frac{\lambda_0^b e^{-1}}{\kappa^b}\exp\left[-\kappa^b(h(t,s,q) - h(t,s,q+1))\right] = 0. \end{aligned} \tag{6}$$

It is difficult to make further analytical progress with (6). In what follows, we propose an approximation method under which the optimal spreads become available in closed form. Similar to Avellaneda and Stoikov (2008), consider the second-order approximation[8]

$$\begin{aligned}\exp\left[-\kappa^a(h(t,s,q) - h(t,s,q-1))\right] &\approx 1 - \kappa^a(h(t,s,q) - h(t,s,q-1)) + \frac{(\kappa^a)^2}{2}(h(t,s,q) - h(t,s,q-1))^2, \\ \exp\left[-\kappa^b(h(t,s,q) - h(t,s,q+1))\right] &\approx 1 - \kappa^b(h(t,s,q) - h(t,s,q+1)) + \frac{(\kappa^b)^2}{2}(h(t,s,q) - h(t,s,q+1))^2. \end{aligned} \tag{7}$$

Then (6) can be approximated by

$$\begin{aligned}\partial_t h &+ \frac{\sigma^2(t,s) - \sigma_{imp}^2}{2}\Gamma^{\$}(t,s)q - \beta q^2 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}h \\ &+ \frac{\lambda_0^a e^{-1}}{\kappa^a}\left[1 - \kappa^a(h(t,s,q) - h(t,s,q-1)) + \frac{(\kappa^a)^2}{2}(h(t,s,q) - h(t,s,q-1))^2\right] \\ &+ \frac{\lambda_0^b e^{-1}}{\kappa^b}\left[1 - \kappa^b(h(t,s,q) - h(t,s,q+1)) + \frac{(\kappa^b)^2}{2}(h(t,s,q) - h(t,s,q+1))^2\right] = 0. \end{aligned} \tag{8}$$

An ansatz to (8) is

$$h(t,s,q) = \psi_0(t,s) + \psi_1(t,s)q + \psi_2(t)q^2$$

[8] In Avellaneda and Stoikov (2008) where symmetric order arrival rate $\lambda_0^a = \lambda_0^b$ and $\kappa^a = \kappa^b$ are assumed, a first-order expansion is used to approximate an exponential term in the HJB equation but such approximation might lose accuracy when $\lambda_0^a \neq \lambda_0^b$ or $\kappa^a \neq \kappa^b$. Hence we adopt a second-order approximation here instead. See also Bergault et al. (2021) who consider a similar quadratic approximation in a problem of multi-asset market making

for some functions ψ_0, ψ_1 and ψ_2 to be determined which terminal conditions are $\psi_0(T,s)=0$, $\psi_1(T,s)=0$ and $\psi_2(T)=-\alpha$. Direct substitution of the ansatz into (8) yields

$$\begin{aligned}
0 = &\Bigg\{\partial_t\psi_0 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}\psi_0 + \frac{\lambda_0^a e^{-1}}{\kappa^a}\left(1-\kappa^a\psi_1+\kappa^a\psi_2+\frac{(\kappa^a)^2}{2}(\psi_1-\psi_2)^2\right) \\
&+ \frac{\lambda_0^b e^{-1}}{\kappa^b}\left(1+\kappa^b\psi_1+\kappa^b\psi_2+\frac{(\kappa^b)^2}{2}(\psi_1+\psi_2)^2\right)\Bigg\} + \Bigg\{\partial_t\psi_1 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}\psi_1 + \frac{\sigma^2(t,s)-\sigma^2_{imp}}{2}\Gamma^{\$}(t,s) \\
&+ 2e^{-1}(\lambda_0^a\kappa^a+\lambda_0^b\kappa^b)\psi_2\psi_1 + 2e^{-1}(\lambda_0^b-\lambda_0^a)\psi_2 + 2e^{-1}(\lambda_0^b\kappa^b-\lambda_0^a\kappa^a)\psi_2^2\Bigg\}q \\
&+ \left\{\partial_t\psi_2 + 2e^{-1}(\lambda_0^b\kappa^b+\lambda_0^a\kappa^a)(\psi_2)^2 - \beta\right\}q^2.
\end{aligned}$$

On equating each coefficient to zero, we obtain the following system of PDEs

$$\begin{cases}
\partial_t\psi_0 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}\psi_0 + \frac{\lambda_0^a e^{-1}}{\kappa^a}\left(1-\kappa^a\psi_1+\kappa^a\psi_2+\frac{(\kappa^a)^2}{2}(\psi_1-\psi_2)^2\right) + \frac{\lambda_0^b e^{-1}}{\kappa^b}\left(1+\kappa^b\psi_1+\kappa^b\psi_2+\frac{(\kappa^b)^2}{2}(\psi_1+\psi_2)^2\right) = 0; \\
\partial_t\psi_1 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}\psi_1 + \frac{\sigma^2(t,s)-\sigma^2_{imp}}{2}\Gamma^{\$}(t,s) + 2e^{-1}(\lambda_0^a\kappa^a+\lambda_0^b\kappa^b)\psi_2\psi_1 + 2e^{-1}(\lambda_0^b-\lambda_0^a)\psi_2 + 2e^{-1}(\lambda_0^b\kappa^b-\lambda_0^a\kappa^a)\psi_2^2 = 0; \\
\partial_t\psi_2 + 2e^{-1}(\lambda_0^b\kappa^b+\lambda_0^a\kappa^a)(\psi_2)^2 - \beta = 0.
\end{cases} \tag{9}$$

The last equation is a simple Riccati ODE with terminal condition $\psi_2(T)=-\alpha$, which solution is

$$\psi_2(t) = \sqrt{\Upsilon\beta}\,\frac{\zeta_- e^{-\eta(T-t)} - \zeta_+ e^{\eta(T-t)}}{\zeta_- e^{-\eta(T-t)} + \zeta_+ e^{\eta(T-t)}},$$

where

$$\Upsilon := \frac{1}{2e^{-1}(\lambda_0^a\kappa^a+\lambda_0^b\kappa^b)}, \qquad \eta := \sqrt{\frac{\beta}{\Upsilon}}, \qquad \zeta_\pm := \sqrt{\Upsilon\beta}\pm\alpha.$$

With the expression of ψ_2, the second equation in (9) is now a linear parabolic PDE (with a source term) in form of

$$\partial_t\psi_1 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}\psi_1 - r(t)\psi_1 + f(t,s) = 0$$

with

$$r(t) := -2e^{-1}(\lambda_0^a\kappa^a+\lambda_0^b\kappa^b)\psi_2(t), \quad f(t,s) := \frac{\sigma^2(t,s)-\sigma^2_{imp}}{2}\Gamma^{\$}(t,s)+2e^{-1}(\lambda_0^b-\lambda_0^a)\psi_2(t)+2e^{-1}(\lambda_0^b\kappa^b-\lambda_0^a\kappa^a)\psi_2^2(t).$$

By the Feynman-Kac representation, the solution admits a probabilistic expression of

$$\psi_1(t,s) = \mathbb{E}\left[\int_t^T \exp\left\{-\int_t^u r(v)dv\right\} f(u,S_u)du \,\Big|\, S_t = s\right].$$

One can compute that

$$\exp\left(-\int_t^u r(v)dv\right) = \exp\left(\int_t^u 2e^{-1}(\lambda_0^a\kappa^a+\lambda_0^b\kappa^b)\psi_2(v)dv\right) = \frac{\zeta_- e^{-\eta(T-u)} + \zeta_+ e^{\eta(T-u)}}{\zeta_- e^{-\eta(T-t)} + \zeta_+ e^{\eta(T-t)}} =: D(t,u). \tag{10}$$

Hence we deduce

$$\psi_1(t,s) = \varphi(t,s) + 2e^{-1}(\lambda_0^b-\lambda_0^a)\int_t^T D(t,u)\psi_2(u)du + 2e^{-1}(\lambda_0^b\kappa^b-\lambda_0^a\kappa^a)\int_t^T D(t,u)\psi_2^2(u)du,$$

where

$$\varphi(t,s) := \mathbb{E}\left[\int_t^T D(t,u)\frac{\sigma^2(u,S_u)-\sigma_{imp}^2}{2}\Gamma^{\$}(u,S_u)du\Big|S_t=s\right]. \tag{11}$$

If we ignore the factor D, φ defined in (11) is precisely the time-t expected total volatility arbitrage profit per unit long position in the option with delta-hedge. Broadly speaking, φ represents the gamma-theta carry of the trader penalised by some "discount factor" linked to the penalty and order flow parameters. In an important special case that the market maker's subjective volatility estimate is a constant $\sigma_t \equiv \sigma$, one can compute $\mathbb{E}\left[\Gamma^{\$}(u,S_u)|S_t=s\right]$ in closed form and in turn obtain

$$\varphi(t,s) = \frac{K(\sigma^2-\sigma_{imp}^2)}{2}\int_t^T \frac{D(t,u)}{\sqrt{\sigma_{imp}^2(\tau-u)+\sigma^2(u-t)}} \times \phi\left(\frac{\ln\frac{s}{K}-\frac{\sigma^2}{2}(u-t)-\frac{\sigma_{imp}^2}{2}(\tau-u)}{\sqrt{\sigma_{imp}^2(\tau-u)+\sigma^2(u-t)}}\right)du, \tag{12}$$

where $\phi(\cdot)$ again is the probability density function of a standard normal random variable. See Ahmad and Wilmott (2005) for the derivation of the above expression.

Knowing the forms of ψ_1 and ψ_2, one can deduce the optimal spreads from (5) as

$$\begin{aligned}
\delta^{a,*} = \delta^{a,*}(t,s,q) &= \frac{1}{\kappa^a} + \psi_1(t,s) + (2q-1)\psi_2(t) \\
&= \frac{1}{\kappa^a} + \varphi(t,s) + 2e^{-1}(\lambda_0^b-\lambda_0^a)\int_t^T D(t,u)\psi_2(u)du + 2e^{-1}(\lambda_0^b\kappa^b-\lambda_0^a\kappa^a)\int_t^T D(t,u)\psi_2^2(u)du \\
&\quad + (2q-1)\psi_2(t)
\end{aligned} \tag{13}$$

and

$$\begin{aligned}
\delta^{b,*} = \delta^{b,*}(t,s,q) &= \frac{1}{\kappa^b} - \psi_1(t,s) - (2q+1)\psi_2(t) \\
&= \frac{1}{\kappa^b} - \varphi(t,s) - 2e^{-1}(\lambda_0^b-\lambda_0^a)\int_t^T D(t,u)\psi_2(u)du - 2e^{-1}(\lambda_0^b\kappa^b-\lambda_0^a\kappa^a)\int_t^T D(t,u)\psi_2^2(u)du \\
&\quad - (2q+1)\psi_2(t).
\end{aligned} \tag{14}$$

The optimal quotes are completely analytical under the case of constant volatility such that the gamma-theta carry term $\varphi(t,s)$ is given by (12), but otherwise φ can still be computed by standard numerical methods conveniently. To gain insights into (13) and (14), we perform Taylor expansions in small $T-t$ (i.e. small market making horizon which is our main application focus) and write approximately $\alpha^2 \approx 0$, we informally have

$$\begin{aligned}
\delta^{a,*} &\approx \frac{1}{\kappa^a} + \varphi(t,s) - \alpha\left(2q-1+2e^{-1}(\lambda_0^b-\lambda_0^a)(T-t)\right) - \beta(2q-1)(T-t), \\
\delta^{b,*} &\approx \frac{1}{\kappa^b} - \varphi(t,s) + \alpha\left(2q+1+2e^{-1}(\lambda_0^b-\lambda_0^a)(T-t)\right) + \beta(2q+1)(T-t).
\end{aligned}$$

Thus the financial interpretation of the optimal spreads can be roughly understood as follows. The first term $1/\kappa^i$ is related to the elasticity of demand and supply of the options. More sensitive the rate of order flow to the quoted spreads (large κ^i) is, more competitive quotes shall be provided. The second component φ represents the "edge" of the market maker in terms of the expected volatility arbitrage profit that they can generate: if for example the trader expects to make money from volatility arbitrage via longing options because they believe the actual volatility will be higher than the implied volatility (i.e. large and positive φ), they are happy with paying a higher price to acquire such position and hence a more competitive (conservative) bid (ask) is quoted. Finally, the last two terms are some adjustment components governed by the penalty parameters α, β, the current inventory level q, the order flow imbalance $\lambda_0^b - \lambda_0^a$ and the remaining duration of the market making horizon $T - t$.

More generally, the term $\varphi(t, s)$ can be viewed as an "alpha signal" which value depends on the trader's view/belief over the subjective volatility relative to the fair implied volatility, and the signal scales with the option gamma over the trajectory of the spot price path. In practice, the market maker can modify the quoted spreads in (13) and (14) via replacing $\varphi(t, s)$ by $\rho\varphi(t, s)$, where $\rho \in [0, 1]$ is a parameter governing how aggressively the alpha signal is utilised. A choice of $\rho = 0$ means the alpha signal is not used at all, which might be due to a very conservative mandate that the market maker is not allowed to quote on the basis of speculation. In contrast, an aggressive trader who is confident with their volatility forecast can consider $\rho = 1$ to fully capitalise the information they have against the market (and such strategy of course is optimal when $\sigma_t = \sigma(t, S_t)$ correctly specifies the true volatility process).

To close this section, we emphasise that the expressions in (13) and (14) are not theoretically optimal due to the second-order approximation used to simplify the HJB equation. We expect them to perform well when the market marking horizon is short and the inventory level is not extremely high or low. In an example in Section 5.1, we will compare the approximated optimal quotes against the "ground truth" obtained from the numerical solutions of the original PDE in (6) where we observe very satisfactory performance. As a final remark, (13) and (14) are also the optimal spreads for a put option with strike K and maturity of τ. Since we assume the market maker fully delta-hedges, call option and put option are financially equivalent. Indeed, the characteristics of the underlying option only enter (13) and (14) via the option's dollar gamma term inside the definition of $\varphi(t, s)$. This observation will allow us to extend the results to general European derivatives in Section 4.2.

4 Miscellaneous extensions

In this section, we discuss several possible extensions of our baseline model in Section 2 and briefly outline how the solution method can be modified. Additional assumptions and/or approximations will be imposed when necessary.

4.1 Option position limit

A position limit is now imposed on the market maker such that Q_t must take value between $[\underline{q}, \overline{q}]$ at all time for some $\underline{q} < 0 < \overline{q}$. We assume $\underline{q} \in \mathbb{Z}_-$ and $\overline{q} \in \mathbb{Z}_+$ are some exogenously given constants. In practice, their values can be inferred from, for example, the total vega limit on the trading book which translates into bounds on the quantity of options. This constraint can be incorporated via a pair of modified intensity functions $\lambda_t^b := 1_{(Q_{t-} < \overline{q})} \lambda^b(\delta_t^b)$ and $\lambda_t^a = 1_{(Q_{t-} > \underline{q})} \lambda^a(\delta_t^a)$. Then limit buy (resp. sell) order from the market maker can no longer be filled due to the zero jump intensity of Y^b (resp. Y^a) when the net inventory reaches $\overline{q}$ (resp. $\underline{q}$).

The value function of the market maker is still given by (3). A straightforward calculation shows that (4) now becomes

$$\partial_t h + \frac{\sigma^2(t,s) - \sigma^2_{imp}}{2}\Gamma^{\$}(t,s)q - \beta q^2 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}h + 1_{\{q>\underline{q}\}} \times \frac{\lambda_0^a e^{-1}}{\kappa^a}\exp\left[-\kappa^a(h(t,s,q) - h(t,s,q-1))\right] + 1_{\{q<\overline{q}\}} \times \frac{\lambda_0^b e^{-1}}{\kappa^b}\exp\left[-\kappa^b(h(t,s,q) - h(t,s,q+1))\right] = 0. \tag{15}$$

This equation is similar to the one studied by Guéant et al. (2013), except that there is an additional source term related to the dollar gamma of the option. We need to make few additional assumptions to proceed. But at the same time we will no longer need to approximate the HJB equation at second-order as we did in Section 3.

Since the market making horizon T is typically short, it is reasonable to assume that the dollar gamma is approximately unchanged over $[0,T]$. We therefore assume that $\Gamma^{\$}(t,s) \approx \Gamma^{\$}(0,S_0) =: \Gamma^{\$}_0$, where S_0 is the initial reference spot price. Similar approximation technique via "freezing" the option greek has also been considered by Baldacci et al. (2021). We further assume a constant subjective volatility process $\sigma_t = \sigma$. Finally, we assume $\kappa := \kappa^a = \kappa^b$, i.e. the elasticity of supply and demand are symmetric. Then (15) becomes

$$\partial_t h + \frac{\sigma^2 - \sigma^2_{imp}}{2}\Gamma^{\$}_0 q - \beta q^2 + \frac{\sigma^2}{2}s^2\partial_{ss}h + 1_{\{q>\underline{q}\}} \times \frac{\lambda_0^a e^{-1}}{\kappa}\exp\left[-\kappa(h(t,s,q) - h(t,s,q-1))\right] + 1_{\{q<\overline{q}\}} \times \frac{\lambda_0^b e^{-1}}{\kappa}\exp\left[-\kappa(h(t,s,q) - h(t,s,q+1))\right] = 0. \tag{16}$$

As the terminal condition $h(T,s,q) = -\alpha q^2$ does not depend on s and the approximating equation (16) only involves s at the second-order term, we conjecture that the solution to (16) does not depend on s and is in form of $h(t,s,q) = h(t,q) = \frac{1}{\kappa}\ln w(t,q)$ for some $w(t,q)$ to be determined with terminal condition $w(T,q) = \exp\left(-\alpha\kappa q^2\right)$. Upon substitution, (16) becomes

$$\partial_t w + \frac{\sigma^2 - \sigma^2_{imp}}{2}\Gamma^{\$}_0 \kappa q w(t,q) - \beta\kappa q^2 w(t,q) + 1_{\{q>\underline{q}\}} \times \lambda_0^a e^{-1} w(t,q-1) + 1_{\{q<\overline{q}\}} \times \lambda_0^b e^{-1} w(t,q+1) = 0. \tag{17}$$

Equation (17) can be interpreted as a system of first order ODEs in t indexed by $q \in \{\underline{q}, \underline{q}-1, ..., 0, ..., \overline{q}-1, \overline{q}\}$, which can be more compactly expressed as

$$\begin{bmatrix} \partial_t w(t,\underline{q}) \\ \partial_t w(t,\underline{q}+1) \\ \partial_t w(t,\underline{q}+2) \\ \vdots \\ \partial_t w(t,\overline{q}-1) \\ \partial_t w(t,\overline{q}) \end{bmatrix} = \begin{bmatrix} \beta\kappa\underline{q}^2 - C_0\kappa\underline{q} & -\lambda_0^b e^{-1} & 0 & 0 & \cdots & \cdots & 0 \\ -\lambda_0^a e^{-1} & \beta\kappa(\underline{q}+1)^2 - C_0\kappa(\underline{q}+1) & -\lambda_0^b e^{-1} & 0 & \cdots & \cdots & 0 \\ 0 & -\lambda_0^a e^{-1} & \beta\kappa(\underline{q}+2)^2 - C_0\kappa(\underline{q}+2) & -\lambda_0^b e^{-1} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & \cdots & 0 & -\lambda_0^a e^{-1} & \beta\kappa(\overline{q}-1)^2 - C_0\kappa(\overline{q}-1) & -\lambda_0^b e^{-1} \\ 0 & \cdots & \cdots & \cdots & 0 & -\lambda_0^a e^{-1} & \beta\kappa\overline{q}^2 - C_0\kappa\overline{q} \end{bmatrix} \begin{bmatrix} w(t,\underline{q}) \\ w(t,\underline{q}+1) \\ w(t,\underline{q}+2) \\ \vdots \\ w(t,\overline{q}-1) \\ w(t,\overline{q}) \end{bmatrix}.$$

In the above, $C_0 := \frac{\sigma^2 - \sigma^2_{imp}}{2}\Gamma^{\$}_0$ represents the volatility arbitrage profit per unit time per unit option. The solution to the system $\boldsymbol{w}(t) := [w(t,\underline{q}), ..., w(t,\overline{q})]^\top$ can be expressed in form of matrix exponential

$$\boldsymbol{w}(t) = \exp(-\mathbf{A}(T-t))\boldsymbol{z}$$

where $\boldsymbol{z} := [e^{-\alpha\kappa\underline{q}^2}, ..., e^{-\alpha\kappa\overline{q}^2}]^\top$ is the terminal condition vector and $\mathbf{A}$ is the coefficients matrix in the above system of ODEs.

Once we have identified $w(t,q)$, the optimal spreads can be found using (5) that

$$\delta^{a,*} = \delta^{a,*}(t,q) = \frac{1}{\kappa}\left(1 + \ln w(t,q) - \ln w(t,q-1)\right), \qquad \delta^{b,*} = \delta^{b,*}(t,q) = \frac{1}{\kappa}\left(1 + \ln w(t,q) - \ln w(t,q+1)\right).$$

This completely characterises the solution to the optimal market making problem.

4.2 Market making of a general European derivative

The analysis also extends easily to market making of a derivative product with generic European payoff structure (e.g. vertical spread, butterfly spread, or even more complicated products such as variance swap which is closely related to a logarithm payoff). The key idea is that, by the Carr-Madan replication formula, any vanilla payoff can be synthesised by a static portfolio of European call options of different strikes. In Section 3, one can see from (13) and (14) that the optimal spread levels depend on the underlying call option only via the expected "discounted" volatility arbitrage profit embedded inside the function φ. When the instrument becomes a portfolio of call options, this quantity is simply replaced by the weighted average of the expected volatility arbitrage profit of each individual option due to the linearity of option gamma.

To make the ideas more concrete, consider a derivative product with maturity τ and a payoff function $g(s)$ which is twice-differentiable (in a generalised sense). Using the Carr-Madan replication formula, there exists a weighting measure $\omega(\cdot)$ such that $g(s) = g(0) + \int_0^\infty (s-K)^+ d\omega(K)$. By linearity of derivative pricing, the mark-to-market value of this product at time t is $\hat{O}_t := g(0) + \int_0^\infty O(t, S_t; K) d\omega(K)$, where $O(t, s; K) := BS_{call}(s, K, \tau - t, \sigma_{imp}(K, \tau))$ is the time-t market value of a European call option with strike K. The market maker quotes bid and ask price on this product at level $\hat{O}_t - \delta_t^b$ and $\hat{O}_t + \delta_t^a$ respectively. $(Y_t^b)_{t\geq 0}$, $(Y_t^a)_{t\geq 0}$ and $(Q_t)_{t\geq 0}$ now represent the cumulative units purchased, cumulative units sold, and the net holding of the product. We again assume the market maker is required to delta-hedge the derivative position. Using linearity of delta, at time t they are required to short $\hat{\Delta}_t := Q_t \int_0^\infty \partial_s O(t, S_t; K) d\omega(K)$ units of the underlying asset.

Following similar analysis in Section 2, the market maker's portfolio value process is found to be

$$dV_t = \left(\int_0^\infty \frac{\sigma^2(t, S_t) - \sigma_{imp}^2(\tau, K)}{2} \Gamma^\$(t, S_t; K) d\omega(K) \right) Q_t dt + \delta_t^a dY_t^a + \delta_t^b dY_t^b,$$

where

$$\Gamma^\$(t, s; K) := \frac{K}{\sigma_{imp}(\tau, K)\sqrt{\tau - t}} \phi \left(\frac{\ln \frac{s}{K} - \frac{\sigma_{imp}^2(\tau, K)}{2}(\tau - t)}{\sigma_{imp}(\tau, K)\sqrt{\tau - t}} \right)$$

is the dollar gamma of a call option with strike K. If we write $C(t, s) := \int_0^\infty \frac{\sigma^2(t,s) - \sigma_{imp}^2(\tau, K)}{2} \Gamma^\$(t, s; K) d\omega(K)$ which represents the instantaneous portfolio volatility arbitrage profit when the spot level is s at time t, the HJB equation for problem (3) can now be expressed as

$$\partial_t h + C(t, s)q - \beta q^2 + \frac{\sigma^2}{2} s^2 \partial_{ss} h + \sup_{\delta^a} \{\lambda^a(\delta^a)[\delta^a + h(t, s, q-1) - h(t, s, q)]\}$$
$$+ \sup_{\delta^b} \left\{\lambda^b(\delta^b)[\delta^b + h(t, s, q+1) - h(t, s, q)]\right\} = 0.$$

The structure of the equation is exactly the same as (4) except we have replaced $\frac{\sigma^2(t,s) - \sigma_{imp}^2(\tau, K)}{2} \Gamma^\$(t, s)$ by $C(t, s)$. By the same second-order expansion arguments in Section 3, one can show that the solution to the approximated equation is $h(t, s, q) = \psi_0(t, s) + \psi_1(t, s)q + \psi_2(t)q^2$ for some unknown functions ψ_0, ψ_1 and ψ_2 to be determined. Using coefficients matching again, system (9) now becomes

$$\begin{cases} \partial_t\psi_0 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}\psi_0 + \frac{\lambda_0^a e^{-1}}{\kappa^a}\left(1-\kappa^a\psi_1+\kappa^a\psi_2+\frac{(\kappa^a)^2}{2}(\psi_1-\psi_2)^2\right) + \frac{\lambda_0^b e^{-1}}{\kappa^b}\left(1+\kappa^b\psi_1+\kappa^b\psi_2+\frac{(\kappa^b)^2}{2}(\psi_1+\psi_2)^2\right) = 0; \\ \partial_t\psi_1 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}\psi_1 + C(t,s) + 2e^{-1}(\lambda_0^a\kappa^a+\lambda_0^b\kappa^b)\psi_2\psi_1 + 2e^{-1}(\lambda_0^b-\lambda_0^a)\psi_2 + 2e^{-1}(\lambda_0^b\kappa^b-\lambda_0^a\kappa^a)\psi_2^2 = 0; \\ \partial_t\psi_2 + 2e^{-1}(\lambda_0^b\kappa^b+\lambda_0^a\kappa^a)(\psi_2)^2 - \beta = 0, \end{cases}$$

and the terminal conditions remain the same as $\psi_0(T,s) = \psi_1(T,s) = 0$ and $\psi_2(T) = -\alpha$. The last equation is the same as in the baseline case where we again have

$$\psi_2(t) = \sqrt{\Upsilon\beta}\frac{\zeta_- e^{-\eta(T-t)} - \zeta_+ e^{\eta(T-t)}}{\zeta_- e^{-\eta(T-t)} + \zeta_+ e^{\eta(T-t)}}$$

with constants Υ, η and $\zeta_\pm$ defined as in Section 3. Meanwhile, the second equation is also structurally similar where we can deduce

$$\psi_1(t,s) = \hat{\varphi}(t,s) + 2e^{-1}(\lambda_0^b-\lambda_0^a)\int_t^T D(t,u)\psi_2(u)du + 2e^{-1}(\lambda_0^b\kappa^b-\lambda_0^a\kappa^a)\int_t^T D(t,u)\psi_2^2(u)du$$

with D being defined in (10), but the gamma-theta carry term is now

$$\begin{aligned} \hat{\varphi}(t,s) :&= \mathbb{E}\left[\int_t^T D(t,u)C(u,S_u)du\middle| S_t = s\right] \\ &= \mathbb{E}\left[\int_t^T D(t,u)\left(\int_0^\infty \frac{\sigma^2(t,s)-\sigma_{imp}^2(\tau,K)}{2}\Gamma^\$(u,S_u;K)d\omega(K)\right)du\middle| S_t = s\right] \\ &= \int_0^\infty \mathbb{E}\left[\int_t^T D(t,u)\frac{\sigma^2(t,s)-\sigma_{imp}^2(\tau,K)}{2}\Gamma^\$(u,S_u;K)du\middle| S_t = s\right]d\omega(K) \\ &= \int_0^\infty \varphi(t,s;K)d\omega(K), \end{aligned}$$

where $\varphi(t,s;K) = \varphi(t,s)$ is given by (11) upon replacing σ_{imp} by $\sigma_{imp}(\tau,K)$ and $\Gamma^\$(u,S_u)$ by $\Gamma^\$(u,S_u;K)$. This is simply the total discounted expected portfolio volatility arbitrage profit between t and T. Moreover, using (12) we can deduce a closed form expression of $\hat{\varphi}(t,s)$ when the subjective volatility is a constant.

Finally, the optimal spread levels are given by

$$\begin{aligned} \delta^{a,*} = &\frac{1}{\kappa^a} + \hat{\varphi}(t,s) + 2e^{-1}(\lambda_0^b-\lambda_0^a)\int_t^T D(t,u)\psi_2(u)du + 2e^{-1}(\lambda_0^b\kappa^b-\lambda_0^a\kappa^a)\int_t^T D(t,u)\psi_2^2(u)du \\ &+ (2q-1)\psi_2(t) \end{aligned} \tag{18}$$

and

$$\begin{aligned} \delta^{b,*} = &\frac{1}{\kappa^b} - \hat{\varphi}(t,s) - 2e^{-1}(\lambda_0^b-\lambda_0^a)\int_t^T D(t,u)\psi_2(u)du - 2e^{-1}(\lambda_0^b\kappa^b-\lambda_0^a\kappa^a)\int_t^T D(t,u)\psi_2^2(u)du \\ &- (2q+1)\psi_2(t). \end{aligned} \tag{19}$$

The expression and financial interpretation of the optimal strategy therefore remain almost unchanged, except that portfolio effect is now taken into the account when evaluating the volatility arbitrage profit reflected by the term $\hat{\varphi}(t,s) := \int_0^\infty \varphi(t,s;K)d\omega(K)$.

4.3 Market making of multiple options on the same underlying

Instead of trading a single call option, now we assume the market maker provides bid and ask quotes on N different call options written on the same underlying asset. For $n \in \{1, ..., N\}$, suppose the n^{th} option has strike K_n and maturity τ_n such that its mark-to-market value at time t is $O_t^n = O^n(t, S_t)$ with $O^n(t, s) := BS_{call}(s, K_n, \tau_n - t, \sigma_{imp}(K_n, \tau_n))$. Again, we assume the market making horizon T is short enough such that $T < \min(\tau_1, ..., \tau_N)$. The bid and ask price quoted for the n^{th} option are $O_t^n - \delta_t^{b,n}$ and $O_t^n + \delta_t^{a,n}$ respectively, where $(\delta_t^{a,n}, \delta_t^{b,n})_{t\geq 0}$ are the bid/ask spread processes for each option chosen by the market maker.

Let $Y_t^{b,n}$ and $Y_t^{a,n}$ respectively be the cumulative units purchased and sold for the n^{th} option by time t. For each $n \in \{1, ..., N\}$ and $i \in \{b, a\}$, we model $(Y_t^{i,n})_{t\geq 0}$ as Cox processes with intensity function $\lambda_t^{i,n} = \lambda_0^{i,n} e^{-\kappa^{i,n}\delta_t^{i,n}}$. The net inventory of the n^{th} option at time t is $Q_t^n := Y_t^{b,n} - Y_t^{a,n}$. Write $\mathbf{Q}_t := [Q_t^1, ..., Q_t^N]^\top$ as the inventory vector at time t.

As before, we require the market maker to delta-hedge such that at time t they are required to short $\Delta_t := \sum_{n=1}^N Q_t^n \partial_s O^n(t, S_t)$ units of the underlying asset. Following the same analysis in Section 2, the market maker's portfolio value process is

$$dV_t = \sum_{n=1}^{N} \left[\frac{\sigma^2(t,s) - \sigma_{imp}^2(K_n, \tau_n)}{2} Q_t^n \Gamma^{\$,n}(t, S_t) dt + \delta_t^{a,n} dY_t^{a,n} + \delta_t^{b,n} dY_t^{b,n} \right],$$

where $\Gamma^{\$,n}$ is the dollar gamma of the n^{th} option defined via $\Gamma^{\$,n}(t,s) := s^2 \partial_{ss} O^n(t,s)$.

In the baseline problem with a single option, penalties in form of $-\alpha Q_T^2$ and $-\beta \int_0^T Q_u^2$ are imposed on the residual and running inventory. When multiple options are involved, there are different ways to implement such penalties. We discuss one idea as follows.

Suppose the market maker wants to summarise the risk of a portfolio of options via multiple measures associated with different "risk factors". They seek to maximise profit while minimising exposure to E number of risk factors, where each of them can be expressed as some linear combination of individual option values. Then the objective function can be formulated as

$$\mathbb{E} \left\{ V_T - \int_0^T \sum_{k=1}^{E} \left[\beta_k \left(\sum_{n=1}^{N} \gamma_n^{(k)} \omega_n Q_u^n \right)^2 \right] du - \sum_{k=1}^{E} \left[\alpha_k \left(\sum_{n=1}^{N} \gamma_n^{(k)} \omega_n Q_T^n \right)^2 \right] \right\}. \tag{20}$$

In the above, $\{\omega_n\}_{n=1,...,N}$ are the risk weights of each option, $\{\gamma_n^{(k)}\}_{n=1,...,N;k=1,...,E}$ define the construction of each risk factor, and $\alpha_k \geq 0$ and $\beta_k \geq 0$ are respectively the terminal and running penalty parameters associated with the k^{th} risk factor.

For example, the entire pool of the option instruments may be partitioned into different groups such that options of similar strike levels and maturities belong to the same group. One can then monitor the total vega across the options from the same group. Formally, suppose the N underlying option instruments are partitioned into E number of buckets such that $\mathcal{B}_1 \cup \mathcal{B}_2 \cup ... \cup \mathcal{B}_E = \{1, 2, ..., N\}$ with $\mathcal{B}_i \cap \mathcal{B}_j = \emptyset$ for all $i \neq j$. Then we say that the n^{th} option belongs to the k^{th} risk bucket if $n \in \mathcal{B}_k$. The residual vega exposure of bucket k is approximated as $\sum_{n\in\mathcal{B}_k} \text{Vega}_0^n Q_T^n$, where Vega_0^n is the initial vega of option n (i.e. we assume the option vega is approximately unchanged over the market making horizon). We can consider a performance criterion (say without running penalty for brevity) of

$$\mathbb{E}\left\{V_T - \sum_{k=1}^{E}\left[\alpha_k\left(\sum_{n\in\mathcal{B}_k}\text{Vega}_0^n Q_T^n\right)^2\right]\right\}. \tag{21}$$

Here, (21) is a special case of (20) under $\omega_n = \text{Vega}_0^n$, $\gamma_n^{(k)} = 1_{\{n\in\mathcal{B}_k\}}$ and $\beta_k = 0$.

As another example, one might wish to protect the options portfolio from structural changes in the implied volatility surface such as parallel shift and movement in skew. Exposure to parallel shift can be captured by the total vega across all instruments, i.e. we choose the risk factor to be $\sum_{n=1}^N \text{Vega}_0^n Q_T^n$. Meanwhile, exposure to skew can be proxied by the vega differential of options on the two ends of the moneyness. An example of an appropriate risk factor can be

$$\sum_{n=1}^{N}(1_{\{K_n\le 0.9S_0\}} - 1_{\{K_n\ge 1.1S_0\}})\text{Vega}_0^n Q_T^n,$$

where K_n is the strike of option n. If one wants to simultaneously penalise exposure to parallel shift as well as change in skewness of the implied volatility surface, then in (20) one can set $E=2$, $\omega_n = \text{Vega}_0^n$, $\gamma_n^{(1)} = 1$ for all n and $\gamma_n^{(2)} = 1_{\{K_n\le 0.9S_0\}} - 1_{\{K_n\ge 1.1S_0\}}$.

Let $\mathbf{R}_\alpha \in \mathbb{R}^{E\times N}$ and $\mathbf{R}_\beta \in \mathbb{R}^{E\times N}$ be matrices which (k,n) elements are given by $[\mathbf{R}_\alpha]_{k,n} := \gamma_n^{(k)}\omega_n\sqrt{\alpha_k}$ and $[\mathbf{R}_\beta]_{k,n} := \gamma_n^{(k)}\omega_n\sqrt{\beta_k}$ respectively. Then the performance goal (20) can be restated as

$$\mathbb{E}\left[V_T - \int_0^T \mathbf{Q}_u^\top \mathbf{B}\mathbf{Q}_u du - \mathbf{Q}_T^\top \mathbf{A}\mathbf{Q}_T\right], \tag{22}$$

where $\mathbf{A} := \mathbf{R}_\alpha^\top \mathbf{R}_\alpha \in \mathbb{R}^{N\times N}$ and $\mathbf{B} := \mathbf{R}_\beta^\top \mathbf{R}_\beta \in \mathbb{R}^{N\times N}$ are symmetric positive definite matrices. Introduce the value function as

$$\begin{aligned} h(t,s,\mathbf{q}) := \sup_{\delta^{i,n}}\mathbb{E}\Bigg[\int_t^T\Bigg(\sum_{n=1}^N\Big[\frac{\sigma^2(t,s)-\sigma^2_{imp}(K_n,\tau_n)}{2}Q_u^n\Gamma^{\$,n}(u,S_u)du + \delta_u^{a,n}dY_u^{a,n} + \delta_u^{b,n}dY_u^{b,n}\Big] - \mathbf{Q}_u^\top\mathbf{B}\mathbf{Q}_u du\Bigg) \\ - \mathbf{Q}_T^\top \mathbf{A}\mathbf{Q}_T \Big| S_t = s, \mathbf{Q}_{t-} = \mathbf{q}\Bigg]. \end{aligned}$$

Then the associated HJB equation is

$$\begin{aligned} \partial_t h + \frac{\sigma^2(t,s)s^2}{2}\partial_{ss}h + \mathbf{C}^\top(t,s)\mathbf{q} - \mathbf{q}^\top\mathbf{B}\mathbf{q} + \sum_{n=1}^N\left\{\sup_{\delta^{a,n}}\lambda^{a,n}(\delta^{a,n})\left[\delta^{a,n} + h\left(t,s,\mathbf{q}-\mathbf{e}_n\right) - h(t,s,\mathbf{q})\right]\right\} \\ + \sum_{n=1}^N\left\{\sup_{\delta^{b,n}}\lambda^{b,n}(\delta^{b,n})\left[\delta^{b,n} + h\left(t,s,\mathbf{q}+\mathbf{e}_n\right) - h(t,s,\mathbf{q})\right]\right\} = 0, \end{aligned}$$

subject to terminal condition $h(T,s,\mathbf{q}) = -\mathbf{q}^\top\mathbf{A}\mathbf{q}$. In the above, $\mathbf{C}(t,s)\in\mathbb{R}^N$ is a vector-valued function such that

$$[\mathbf{C}(t,s)]_n := \frac{\sigma^2(t,s)-\sigma^2_{imp}(K_n,\tau_n)}{2}\Gamma^{\$,n}(t,s)$$

is the volatility arbitrage profit rate per unit option n, and each $\mathbf{e}_n\in\mathbb{R}^N$ is a basis vector (i.e. a vector with value of unity for its n^{th} entry and zero elsewhere). Under $\lambda^{i,n}(\delta) = \lambda_0^{i,n}e^{-\kappa^{i,n}\delta}$, the optimal controls are

$$\delta^{a,n,*} = \frac{1}{\kappa^{a,n}} + h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}-\mathbf{e}_n), \qquad \delta^{b,n,*} = \frac{1}{\kappa^{b,n}} + h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}+\mathbf{e}_n). \tag{23}$$

After substituting (23) back to the HJB equation, we obtain

$$\partial_t h + \frac{\sigma^2(t,s)s^2}{2}\partial_{ss}h + \mathbf{C}^\top(t,s)\mathbf{q} - \mathbf{q}^\top\mathbf{B}\mathbf{q} + \sum_{n=1}^N \frac{\lambda_0^{a,n}e^{-1}}{\kappa^{a,n}}\exp\left(-\kappa^{a,n}(h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}-\mathbf{e}_n))\right)$$
$$+ \sum_{n=1}^N \frac{\lambda_0^{b,n}e^{-1}}{\kappa^{b,n}}\exp\left(-\kappa^{b,n}(h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}+\mathbf{e}_n))\right) = 0.$$

We proceed similarly as in Section 3, where we invoke a second-order approximation of

$$\exp\left(-\kappa^{a,n}(h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}-\mathbf{e}_n))\right) \approx 1 - \kappa^{a,n}(h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}-\mathbf{e}_n)) + \frac{(\kappa^{a,n})^2}{2}(h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}-\mathbf{e}_n))^2,$$
$$\exp\left(-\kappa^{b,n}(h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}+\mathbf{e}_n))\right) \approx 1 - \kappa^{b,n}(h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}+\mathbf{e}_n)) + \frac{(\kappa^{b,n})^2}{2}(h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}+\mathbf{e}_n))^2,$$

for each n. The approximated PDE is now

$$\begin{aligned}
&\partial_t h + \frac{\sigma^2(t,s)s^2}{2}\partial_{ss}h + \mathbf{C}^\top(t,s)\mathbf{q} - \mathbf{q}^\top\mathbf{B}\mathbf{q} \\
&\quad + \sum_{n=1}^N \frac{\lambda_0^{a,n}e^{-1}}{\kappa^{a,n}}\left[1 - \kappa^{a,n}(h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}-\mathbf{e}_n) + \frac{(\kappa^{a,n})^2}{2}(h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}-\mathbf{e}_n))^2\right] \\
&\quad + \sum_{n=1}^N \frac{\lambda_0^{b,n}e^{-1}}{\kappa^{b,n}}\left[1 - \kappa^{b,n}(h(t,s,\mathbf{q})) - h(t,s,\mathbf{q}+\mathbf{e}_n) + \frac{(\kappa^{b,n})^2}{2}(h(t,s,\mathbf{q}) - h(t,s,\mathbf{q}+\mathbf{e}_n))^2\right] = 0,
\end{aligned} \tag{24}$$

with terminal condition $h(T,s,\mathbf{q}) = -\mathbf{q}^\top\mathbf{A}\mathbf{q}$.

Make an ansatz in form of

$$h(t,s,\mathbf{q}) = \theta_0(t,s) + \boldsymbol{\theta}_1^\top(t,s)\mathbf{q} + \mathbf{q}^\top\boldsymbol{\Theta}_2(t)\mathbf{q},$$

where $\theta_0(t,s) \in \mathbb{R}$, $\boldsymbol{\theta}_1(t,s) \in \mathbb{R}^N$, and $\boldsymbol{\Theta}_2(t) \in \mathbb{R}^{N\times N}$ are some (scalar-, vector-, and symmetric matrix-valued) functions to be determined with terminal conditions $\theta_0(T,s) = 0$, $\boldsymbol{\theta}_1(T,s) = \mathbf{0}$ and $\boldsymbol{\Theta}_2(T) = -\mathbf{A}$. Substituting the ansatz into (24) and then equating the 0^{th}, 1^{st} and 2^{nd} order coefficient of $\mathbf{q}$ to be zero, we obtain the following system of PDEs

$$\begin{cases}
\partial_t\theta_0 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}\theta_0 + e^{-1}\Big\{\mathbf{1}^\top(\frac{\boldsymbol{\lambda}_0^a}{\boldsymbol{\kappa}^a} + \frac{\boldsymbol{\lambda}_0^b}{\boldsymbol{\kappa}^b}) + \boldsymbol{\theta}_t^\top(\boldsymbol{\lambda}_0^b - \boldsymbol{\lambda}_0^a) + \mathrm{tr}\Big(diag(\boldsymbol{\lambda}_0^b + \boldsymbol{\lambda}_0^a)diag(\boldsymbol{\Theta}_2)\Big) \\
\qquad + \boldsymbol{\theta}_1^\top diag(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b + \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a)\boldsymbol{\theta}_1 + \mathrm{tr}\Big(diag(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b - \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a)diag(\boldsymbol{\Theta}_2)\boldsymbol{\theta}_1\Big) \\
\qquad + \frac{1}{2}\mathrm{tr}\Big(diag(\boldsymbol{\Theta}_2)diag(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b + \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a)diag(\boldsymbol{\Theta}_2)\Big)\Big\} = 0; \\
\partial_t\boldsymbol{\theta}_1 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}\boldsymbol{\theta}_1 + \mathbf{C}(t,s) + 2e^{-1}\boldsymbol{\Theta}_2 diag(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b + \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a)\boldsymbol{\theta}_1 \\
\qquad + 2e^{-1}\boldsymbol{\Theta}_2(\boldsymbol{\lambda}_0^b - \boldsymbol{\lambda}_0^a) + 2e^{-1}\boldsymbol{\Theta}_2 diag(\boldsymbol{\Theta}_2)(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b - \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a) = \mathbf{0}; \\
\partial_t\boldsymbol{\Theta}_2 - \mathbf{B} + 2e^{-1}\boldsymbol{\Theta}_2 diag(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b + \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a)\boldsymbol{\Theta}_2 = \mathbf{0},
\end{cases} \tag{25}$$

where $\boldsymbol{\lambda}_0^i := [\lambda_0^{i,1}, \lambda_0^{i,2}, ..., \lambda_0^{i,N}]^\top$, $\boldsymbol{\kappa}^i := [\kappa^{i,1}, \kappa^{i,2}, ..., \kappa^{i,N}]^\top$ and $\boldsymbol{\lambda}_0^i/\boldsymbol{\kappa}^i := [\lambda_0^{i,1}/\kappa^{i,1}, \lambda_0^{i,2}/\kappa^{i,2}, ..., \lambda_0^{i,N}/\kappa^{i,N}]^\top$ for $i \in \{b,a\}$; $\odot$ denotes the Hadamard product of two vectors; $\mathbf{1}$ denotes a column vector with unity entries; and $diag(\mathbf{M})$ represents an N-by-N diagonal matrix which diagonal entries are either $(M_1, ..., M_N)$ if $\mathbf{M} \in \mathbb{R}^N$, or

$(M_{11}, ..., M_{NN})$ if $\mathbf{M} \in \mathbb{R}^{N\times N}$. The last equation in (25) is a standard Riccati ODE which solution can be analytically expressed as

$$\mathbf{\Theta}_2(t) = -\frac{1}{2}\mathbf{Y}(t)\mathbf{X}^{-1}(t),$$

where $\mathbf{X}(t) \in \mathbb{R}^{N\times N}$ and $\mathbf{Y}(t) \in \mathbb{R}^{N\times N}$ are solutions to a matrix-valued first order ODE

$$\begin{bmatrix} \partial_t \mathbf{X}(t) \\ \partial_t \mathbf{Y}(t) \end{bmatrix} = \mathbf{H} \begin{bmatrix} \mathbf{X}(t) \\ \mathbf{Y}(t) \end{bmatrix}, \qquad \mathbf{X}(T) = \mathbf{I}_{N\times N}, \quad \mathbf{Y}(T) = 2\mathbf{A}, \tag{26}$$

with Hamiltonian matrix

$$\mathbf{H} := \begin{bmatrix} \mathbf{0}_{N\times N} & -e^{-1}diag(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b + \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a) \\ -2\mathbf{B} & \mathbf{0}_{N\times N} \end{bmatrix} \in \mathbb{R}^{2N\times 2N}.$$

In the above, $\mathbf{0}_{N\times N}$ and $\mathbf{I}_{N\times N}$ denote the N-by-N zero matrix and identity matrix respectively. After we have solved for $\mathbf{\Theta}_2(t)$, the second equation in (24) is a coupled system of linear parabolic PDEs. Analytical solutions do not appear to be viable in general, but standard numerical methods such as explicit finite difference can still be employed to solve for $\boldsymbol{\theta}_1(t,s)$ quite efficiently.

Nonetheless, under further assumption and approximation, it is indeed possible to obtain semi-explicit solutions of $\mathbf{\Theta}_2$ and $\boldsymbol{\theta}_1$. Assume $\mathbf{B} = \mathbf{0}_{N\times N}$ which arises for example when there is no running penalty (i.e. $\beta_k = 0$ for all $k \in \{1, ..., E\}$.) In this case, $\mathbf{Y}(t) = 2\mathbf{A}$ and $\mathbf{X}(t) = \mathbf{I} + 2e^{-1}(T-t)diag(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b + \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a)\mathbf{A}$. Then we obtain the closed form solution of $\mathbf{\Theta}_2$ as

$$\mathbf{\Theta}_2(t) = -\frac{1}{2}\mathbf{Y}(t)\mathbf{X}^{-1}(t) = -\mathbf{A}(\mathbf{I} + 2e^{-1}(T-t)diag(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b + \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a)\mathbf{A})^{-1}. \tag{27}$$

If one further considers the specialisation of constant subjective volatility and an approximation of constant dollar gamma same as we did in Section 4.1, then $[\mathbf{C}(t,s)]_n = \frac{\sigma^2 - \sigma^2_{imp}(K_n,\tau_n)}{2}\Gamma^{\$,n}(t,s) \approx \frac{\sigma^2 - \sigma^2_{imp}(K_n,\tau_n)}{2}\Gamma^{\$,n}(0,S_0) =:$ $[\mathbf{C}_0]_n$ for all t and s. Here, $\mathbf{C}_0 \in \mathbb{R}^N$ is a constant vector collecting the estimated expected volatility arbitrage rate of each option. Under this approximation, $\boldsymbol{\theta}_1(t,s)$ no longer depends on s and the system of PDEs becomes

$$\partial_t \boldsymbol{\theta}_1 + \mathbf{C}_0 + 2e^{-1}\mathbf{\Theta}_2 diag(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b + \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a)\boldsymbol{\theta}_1 + 2e^{-1}\mathbf{\Theta}_2(\boldsymbol{\lambda}_0^b - \boldsymbol{\lambda}_0^a) + 2e^{-1}\mathbf{\Theta}_2 diag(\mathbf{\Theta}_2)(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b - \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a) = \mathbf{0}.$$

This is simply a system of first order linear ODEs in form of

$$\partial_t \boldsymbol{\theta}_1(t) + \mathbf{\Phi}(t)\boldsymbol{\theta}_1(t) + \mathbf{f}(t) = \mathbf{0}, \qquad \boldsymbol{\theta}_1(T) = \mathbf{0},$$

with $\mathbf{\Phi}(t) := 2e^{-1}\mathbf{\Theta}_2(t)diag(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b + \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a)$ and $\mathbf{f}(t) := 2e^{-1}\mathbf{\Theta}_2(t)(\boldsymbol{\lambda}_0^b - \boldsymbol{\lambda}_0^a) + 2e^{-1}\mathbf{\Theta}_2(t)diag(\mathbf{\Theta}_2(t))(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b - \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a) + \mathbf{C}_0$. As we work with $\mathbf{B} = 0$ where $\mathbf{\Theta}_2$ is given by (27), we have

$$\begin{aligned} \mathbf{\Xi}(t,u) := \int_t^u \mathbf{\Phi}(v)dv &= 2e^{-1}\left(\int_t^u \mathbf{\Theta}_2(v)dv\right)\mathbf{D} \\ &= -2e^{-1}\left(\int_t^u \mathbf{A}(\mathbf{I} + 2e^{-1}(T-v)\mathbf{D}\mathbf{A})^{-1}dv\right)\mathbf{D} \\ &= \mathbf{A}\left[\ln\big(\mathbf{I} + 2e^{-1}(T-u)\mathbf{D}\mathbf{A}\big) - \ln\big(\mathbf{I} + 2e^{-1}(T-t)\mathbf{D}\mathbf{A}\big)\right]\mathbf{A}^{-1}, \end{aligned}$$

where we have defined $\mathbf{D} := diag(\boldsymbol{\lambda}_0^b \odot \boldsymbol{\kappa}^b + \boldsymbol{\lambda}_0^a \odot \boldsymbol{\kappa}^a)$ for brevity and the operator $\ln(\cdot)$ above refers to matrix logarithm.[9] In turn, we can explicitly express the solution with the help of integrating factor as[10]

$$\begin{aligned}\boldsymbol{\theta}_1(t) = \int_t^T e^{\boldsymbol{\Xi}(t,u)}\mathbf{f}(u)du &= \int_t^T (\mathbf{I} + 2e^{-1}(T-t)\mathbf{AD})^{-1}(\mathbf{I} + 2e^{-1}(T-u)\mathbf{AD})\mathbf{f}(u)du \\ &= (\mathbf{I} + 2e^{-1}(T-t)\mathbf{AD})^{-1}\int_t^T (\mathbf{I} + 2e^{-1}(T-u)\mathbf{AD})\mathbf{f}(u)du. \qquad (28)\end{aligned}$$

Regardless of the simplifying approximation/assumptions used, once we know $\boldsymbol{\Theta}_2(t)$ and $\boldsymbol{\theta}_1(t,s)$ we can use (23) to deduce the optimal spreads as

$$\delta^{a,n,*} = \frac{1}{\kappa^{a,n}} + \boldsymbol{\theta}_1^\top(t,s)\mathbf{e}_n - \mathbf{e}_n^\top \boldsymbol{\Theta}_2(t)\mathbf{e}_n + 2\mathbf{e}_n^\top \boldsymbol{\Theta}_2(t)\mathbf{q}, \qquad \delta^{b,n,*} = \frac{1}{\kappa^{b,n}} - \boldsymbol{\theta}_1^\top(t,s)\mathbf{e}_n - \mathbf{e}_n^\top \boldsymbol{\Theta}_2(t)\mathbf{e}_n - 2\mathbf{e}_n^\top \boldsymbol{\Theta}_2(t)\mathbf{q},$$

which can be more compactly represented in vector-form

$$\boldsymbol{\delta}^{a,*}(t,s,\mathbf{q}) = \frac{1}{\boldsymbol{\kappa}^a} + \boldsymbol{\theta}_1(t,s) - diag(\boldsymbol{\Theta}_2(t))\mathbf{1} + 2\boldsymbol{\Theta}_2(t)\mathbf{q}; \qquad (29)$$

$$\boldsymbol{\delta}^{b,*}(t,s,\mathbf{q}) = \frac{1}{\boldsymbol{\kappa}^b} - \boldsymbol{\theta}_1(t,s) - diag(\boldsymbol{\Theta}_2(t))\mathbf{1} - 2\boldsymbol{\Theta}_2(t)\mathbf{q}. \qquad (30)$$

Note that the optimal market making rules for option n in generally depend on the state value of the entire inventory vector $\mathbf{q} = [q^1, q^2, ..., q^n]^\top$, rather than just the inventory level of that individual option q^n. This is due to the fact that portfolio effect is now taken into the account based on the construction of the underlying risk factors.

5 Numerical illustrations

We present a few numerical examples to demonstrate the market making algorithms presented in the previous sections.

5.1 Market making of a single European call option

In the first example, we consider the market making of a single one-month at-the-money call option ($\tau = 1/12$ and $K = S_0$). Initial spot price and implied volatility are chosen to be $S_0 = 100$ and $\sigma_{imp} = 30\%$. The market making horizon is $T = 0.5/252$, representing about half a trading day. We take $\lambda_0^a = \lambda_0^b = 50 \times 252$, i.e. on each trading day 50 buy and 50 sell orders will be executed on average if prices are quoted at the "fair mid prices" with zero spreads. Set $\kappa^a = \kappa^b = 0.75/(1\% \times \text{Vega}_0)$ where Vega_0 is the initial vega of the option. Then roughly speaking when the quoted price increases by 1% implied volatility point away from the mid price, the average number of buy/sell order flow will decrease by a factor of $e^{-0.75} \approx 0.47$. Finally, we take $\alpha = 0.01 \times (1\% \times \text{Vega}_0)^2$ and $\beta = 0.005 \times (1\% \times \text{Vega}_0)^2$, under which penalty multipliers of 0.01 and 0.005 are applied to percentage-vega of the residual and running option inventory respectively.

[9] Note that $\mathbf{X}(v) := \mathbf{I} + 2e^{-1}(T-v)\mathbf{DA}$ has strictly positive real eigenvalues for any $v \in [0,T]$ since $\mathbf{A}$ is positive semidefinite. Moreover, $\mathbf{X}(v)$ commutes with $\mathbf{X}'(v)$. Then $\frac{d}{dv}\ln\mathbf{X}(v) = \mathbf{X}'(v)\mathbf{X}^{-1}(v) = \mathbf{X}^{-1}(v)\mathbf{X}'(v)$ and $\ln\mathbf{X}^{-1}(v) = -\ln\mathbf{X}(v)$.

[10] One can verify that the matrices $\boldsymbol{\Xi} = \int \boldsymbol{\Phi}(v)dv$ and $\boldsymbol{\Phi}$ are commutative and hence the solution to the system of ODE can be solved by the method of integrating factor.

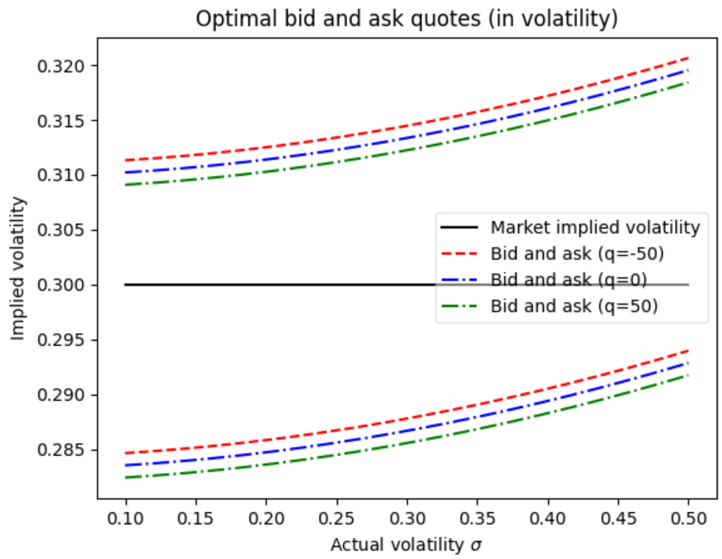
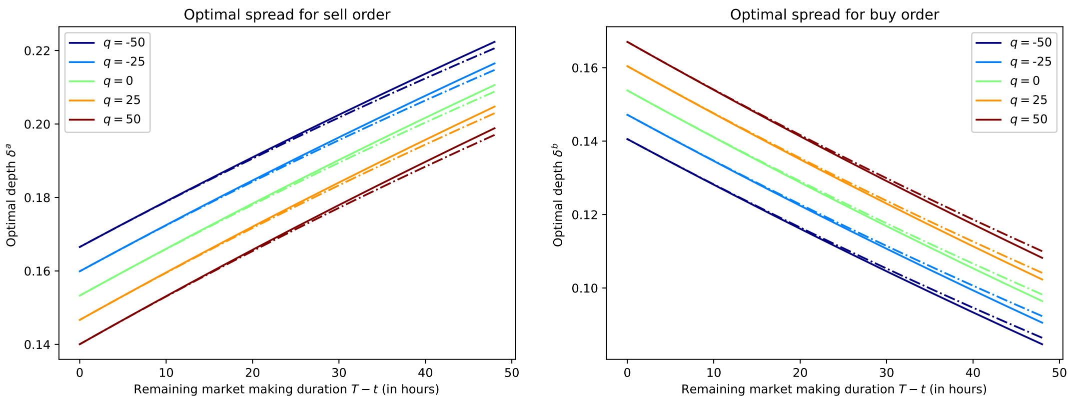


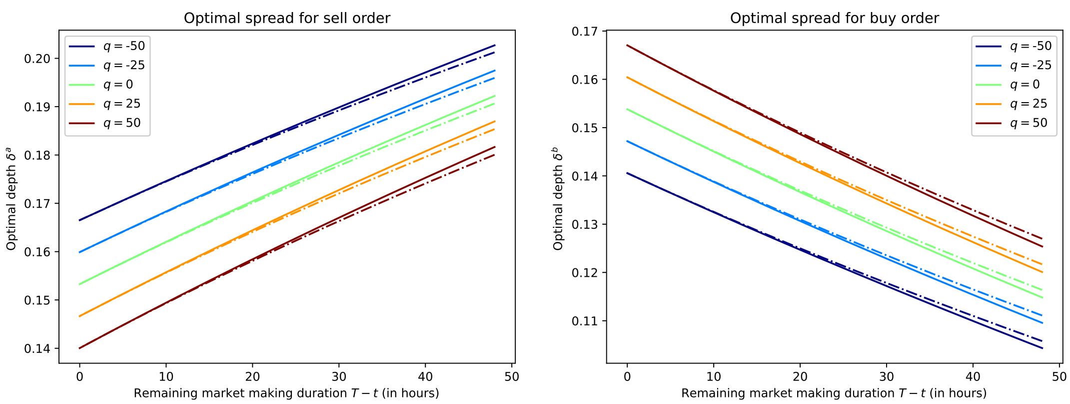
Figure 1: Optimal quotes on a European call option computed by equations (13) and (14).

In Figure 1, we use formulae (13) and (14) in conjunction with (12) to compute the optimal bid and ask prices under constant subjective volatility assumption. We express the quotes in terms of implied volatility level, and we focus on the optimal quotes at time zero under different values of σ the actual volatility estimation and q the current inventory level. With a higher actual volatility σ, both the bid and ask quotes will skew upwards such that the market maker provides a more competitive (resp. conservative) bid (resp. ask) price. It is because a large and positive actual-versus-implied volatility differential makes long positions of the call option profitable. The market maker is therefore happy with accumulating more inventory since the lost profit due to a tighter spread can be compensated by a larger volatility arbitrage profit. Figure 1 also shows that the quotes are decreasing in the current inventory level. Under a large and positive inventory level, the market maker is incentivised to offload their inventory via providing a lower ask (resp. bid) price to increase (resp. decrease) the chance of option sale (resp. purchase), as to reduce the penalty imposed on the residual/running inventory.

To illustrate the quality of our optimal quotes under the second-order approximation of the HJB equation, we also compute the quotes via numerically solving the original PDE in (6) by a finite difference scheme with operator splitting (backward in the linear component and forward in the non-linear component). The model parameters used are the same as before expect the market making horizon is increased to $T = 2/252$ (two business days) such that we can compare the quotes over a longer duration, and we fix the actual volatility to be $\sigma = 0.35$. As shown in Figure 2, the differences between the quotes obtained by the two different methods are numerically and economically insignificant (within 0.015% annualised implied volatility), and are only distinguishable when the market making duration extends beyond an entire business day. The performance is also robust to change in the model parameters such as an increase in order flow imbalance.



(a) Equal buy and sell baseline order arrival rates $\lambda_0^a = \lambda_0^b = 50 \times 252$.



(b) Unequal buy and sell baseline order arrival rates $\lambda_0^a = 50 \times 252$ and $\lambda_0^b = 150 \times 252$.

Figure 2: Comparison of the optimal quotes for the call option obtained by formulae in (13) and (14) (indicated by the dashed lines) versus those obtained by numerical solutions to the original PDE in (6) (indicated by the solid lines). Actual volatility is fixed at $\sigma = 0.35$.

5.2 Market making of a risk reversal

Next, we look at the market making of a six-month 95-105 risk reversal consisting a long position of a six-month call of 105% moneyness and a short position of a six-month put of 95% moneyness. The spot is fixed at $S_0 = 100$. The call and the put option then have strike of $K_1 = 105$ and $K_2 = 95$ respectively, and a common maturity of $\tau = 6/12$. The baseline market implied volatility of the two options are assumed to be $\sigma_{imp}(K_1, \tau) = 12\%$ and $\sigma_{imp}(K_2, \tau) = 17\%$. Set $\sigma = 15\%$ to be the actual volatility. We pick $\lambda_0^a = \lambda_0^b = 50 \times 252$ and $\kappa^a = \kappa^b = 0.75/(1\% \times \overline{\text{Vega}_0})$ where $\overline{\text{Vega}_0}$ is average initial vega of the call and the put option. α is chosen as $0.01 \times (1\% \times \text{Vega}_0^{RR})^2$ where Vega_0^{RR} is the net initial vega of the risk reversal (i.e. the difference between the call option vega and the put option vega) and we take $\beta = 0$. The market making horizon is $T = 0.5/252$ or half a business day.

We investigate how the optimal posted spreads (at time zero under zero initial inventory) are affected by shocks to the market implied volatility surface. We consider two types of shock: parallel shock and skew shock. In the former case, we look at the optimal quotes under implied volatility values of $\sigma_{imp}(K_1, \tau) = 12\% + \Delta_{\text{parallel}}$ and $\sigma_{imp}(K_2, \tau) = 17\% + \Delta_{\text{parallel}}$ as Δ_{parallel} varies. In the latter case, we consider implied volatility levels of $\sigma_{imp}(K_1, \tau) = 12\% + \Delta_{\text{skew}}$ and $\sigma_{imp}(K_2, \tau) = 17\% - \Delta_{\text{skew}}$ under different values of Δ_{skew}.[11] The optimal spreads are computed by formulae (18) and (19), in which the volatility arbitrage term $\hat{\phi}$ is given by

$$\hat{\varphi}(t, s) = \varphi(t, s; K_1) - \varphi(t, s; K_2).$$

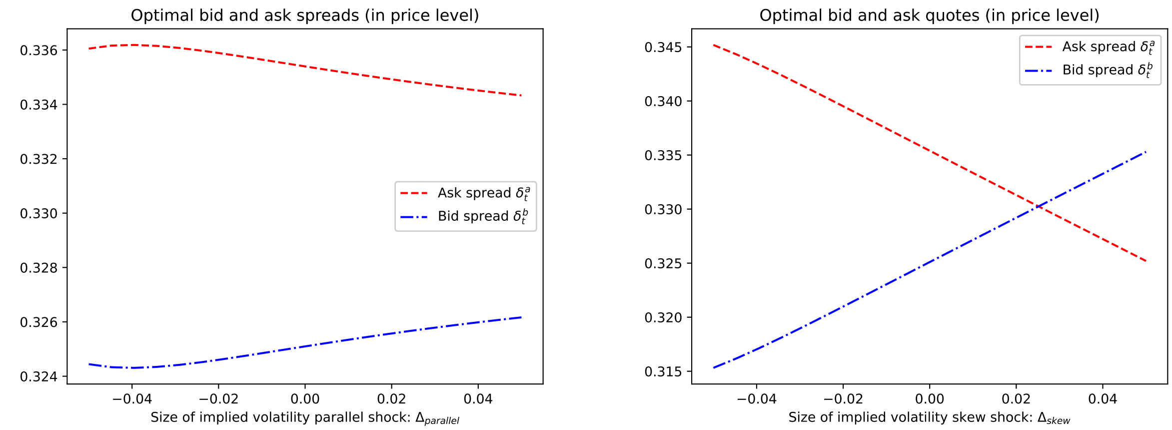
Here $\varphi(t, s; K_1)$ and $\varphi(t, s; K_2)$ denote the total expected "discounted" volatility arbitrage associated with the call and the put respectively, as given by (12).

The results are shown in Figure 3 where the optimal bid and ask spreads are stated in price level. We observe that change of the implied volatility skew has a more significant impact on the option quotes, relative to parallel shift of the implied volatility level. It is not surprising because a long position in a (delta-hedged) risk reversal is equivalent to shorting the implied volatility skew. An increase in implied volatility skew (i.e. when Δ_{skew} becomes more negative) thus makes it more attractive to long a risk reversal and the market maker will hence shift their quotes upwards (in form of a smaller bid spread and larger ask spread). In contrast, a parallel increase in the implied volatility surface makes the long (resp. short) position in the call (resp. put) option less (resp. more) attractive. These two factors cancel out and the resulting quoted spreads are therefore not significantly affected by the parallel shift.

5.3 Market making of multiple options subject to vega bucket monitoring

As our final example, we consider a market maker overseeing multiple vanilla options of different strikes and expiries on the same underlying. We assume the available expiries are three-month, six-month, one-year and three-year (i.e. $\tau \in \{3/12, 6/12, 1, 3\}$) and the available moneynesses range from 90% to 110% in an increment of 2% (i.e. $\tilde{K} := K/S_0 \in \{0.9, 0.92, 0.94, ..., 1.08, 1.1\}$). Hence there are in total $N = 44$ different option instruments. Note that under our setting it does not matter whether each option is call or put.

[11] For simplicity, we do not change the values of κ^a and κ^b as Δ_{parallel} and Δ_{skew} vary.



(a) Parallel shock.

(b) Skew shock.

Figure 3: Optimal bid and ask spreads (in price level) on a six-month European risk reversal at time zero under zero initial inventory. Implied volatility levels for the call and put leg are in form of $\sigma_{imp}(K_1, \tau) = 12\% + \Delta_{\text{parallel}}$ and $\sigma_{imp}(K_2, \tau) = 17\% + \Delta_{\text{parallel}}$ in panel (a), or $\sigma_{imp}(K_1, \tau) = 12\% + \Delta_{\text{skew}}$ and $\sigma_{imp}(K_2, \tau) = 17\% - \Delta_{\text{skew}}$ under panel (b).

		Range of moneyness $\tilde{K} := K/S_0$		
		$(0, 0.95]$	$(0.95, 1.05)$	$[1.05, \infty)$
Range of maturity τ	$(0, 9/12)$	Bucket 1	Bucket 2	Bucket 3
	$[9/12, \infty)$	Bucket 4	Bucket 5	Bucket 6

Table 1: Options are partitioned into 6 groups (buckets) based on their maturities (long/short) and moneynesses (low/moderate/high).

Strike K	Maturity τ			
	0.25	0.5	1	3
90	0.246070	0.226079	0.211225	0.195586
92	0.231669	0.215332	0.203405	0.191095
94	0.216930	0.204513	0.195638	0.186694
96	0.201910	0.193689	0.187970	0.182395
98	0.186787	0.182981	0.180462	0.178211
100	0.172009	0.172606	0.173205	0.174159
102	0.158501	0.162905	0.166318	0.170258
104	0.147704	0.154350	0.159950	0.166527
106	0.140853	0.147444	0.154264	0.162989
108	0.137922	0.142517	0.149411	0.159666
110	0.137856	0.139552	0.145493	0.156577

Table 2: Implied volatility surface used for the numerical example. Values in each sub-table correspond to the implied volatilities of options from the same bucket.

Suppose the risk monitoring of the options portfolio is performed via bucketing. In particular, options of similar strikes and maturities are grouped together and separate risk penalties will be applied to different buckets. Following the ideas in Section 4.3, we construct $E = 6$ buckets based on the rules shown in Table 1. For each bucket, its risk measure is chosen to be the total percentage-vega of all instruments in the group. Under the approximation that each option's terminal vega is roughly equal to its initial vega, the penalty term for Bucket k can be written as $\alpha_k \left(\sum_{n\in\mathcal{B}_k} Q_T^n \times 1\% \times \text{Vega}_0^n\right)^2$, where $\mathcal{B}_k$ denotes the collection of options in Bucket k and Vega_0^n is the initial vega of an individual option. As a baseline, we pick $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6\} = \{0.2, 0.01, 0.2, 0.2, 0.01, 0.2\}$ such that Bucket 1, 3, 4 and 6 receive a larger risk penalty as to discourage vega exposure at the more extreme level of strike. We will investigate how the market maker's profit and risk change as the α_k's vary. Meanwhile, all running penalty parameters β_k's are set to zero.

Initial spot is chosen to be $S_0 = 100$ while the actual volatility is fixed at $\sigma = 17\%$. The market implied volatility surface is given by Table 2. Regarding the order arrival parameters, we specify $\lambda_0^{a,n} = 252\times 100\times(1-|1-K_n/S_0|)$ for all n. It corresponds to an average of 100 market sale orders executed by the market maker per day for each at-the-money option if the prices are quoted at the market fair level, but this number decreases for options with extreme strikes since they are less liquid. Similarly, we assume $\kappa^{i,n} = 5\times(1-5\times|1-K_n/S_0|)/(1\%\times\text{Vega}_0^n)$ for all n and $i\in\{b,a\}$. The value of $\kappa^{i,n}$ becomes smaller for option with extreme strikes, capturing the idea that the order flow of those options is less sensitive to widening spread (in volatility terms). We take $\lambda_0^{b,n} = 1.15\lambda_0^{a,n}$ for all n, stipulating a scenario with net selling pressure from the market. The market making horizon is again half a trading day such that $T = 0.5/252$. The optimal quotes are computed by formulae (29) and (30), where we set $\omega_n = 1\%\times\text{Vega}_0^n$ and $\gamma_n^{(k)} = 1_{\{n\in\mathcal{B}_k\}}$ for all n and k.

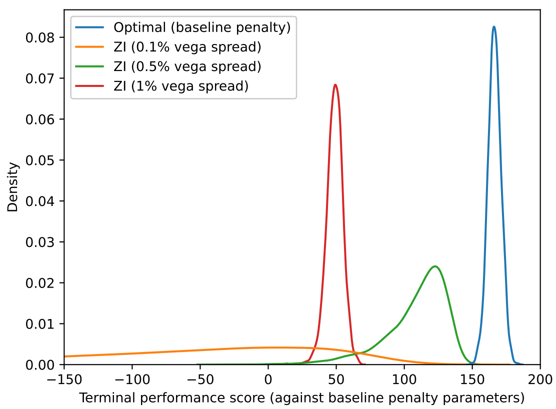
We conduct simulation and look at the trading profit as well as the terminal portfolio vega. We adopt time-discretisation with step size of $\Delta t = 0.25/(60 \times 24 \times 252)$, i.e. the market maker refreshes their quotes and rebalances their delta-hedge every 15 seconds. To benchmark the performance of our model, we further consider three "zero-intelligence" market makers of different aggressiveness who provide quotes using a constant bid and ask spread across all instruments. Three levels of constant spread are considered for each zero-intelligence market maker: 0.1, 0.5 and 1 unit of percentage-vega. 2000 simulations are used. In Figure 4, we plot the empirical densities (kernel density estimates) of the simulated performance scores defined by (22), the terminal portfolio values and the aggregate terminal percentage-vega, for all types of market making strategies. Meanwhile, Figure 5 shows the empirical densities of the total terminal percentage-vega of instruments associated with each of the six risk buckets.

Panel (a) of Figure 4 shows that the proposed algorithmic trading rule outperforms the zero-intelligence strategies in terms of average performance score, which is expected by the design of the optimal strategy. This translates to higher average portfolio values (Panel (b) of Figure 4) and terminal vega values of tighter dispersion (Panel (c) of Figure 4) with the optimal strategy. The zero-intelligence strategy with 1 unit of percentage-vega spread is doing poorly in terms of PnL because the quotes are too conservative to attract enough transactions. The zero-intelligence strategies with 0.1 and 0.5 unit of percentage-vega spread achieve a relatively high PnL but they tend to build up a large positive and volatile vega exposure at the maturity (as the market is a net seller of volatility under our choice of order flow parameters), leading to worse average performance scores.

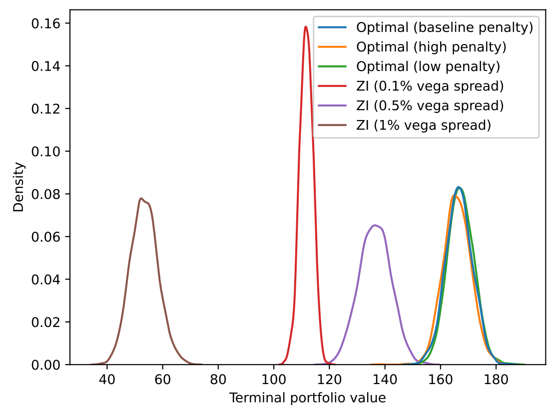
More granular behaviours of the bucket vega are shown in Figure 5. Under the optimal strategy, the sub-portfolios in Bucket 1, 3, 4 and 6 have a relatively narrower range of terminal vega centred around zero whereas those in Bucket 2 and 5 have wider dispersion. It is because we have imposed a more restrictive control over vega risk for out-of-money options through our choice of the baseline α_k's. Modifying the values of the penalty parameters can result in different risk-return profiles attained by the proposed algorithmic market making rule. In Panel (b) of Figure 4, we observe a (very mild) downward shift of the PnL distribution as the penalty parameters increase. Large penalty parameters also lead to a terminal portfolio vega distribution centring around zero tightly, as shown in Panel (c) of Figure 4. In summary, this suggests that our algorithm can deliver superior return with more flexible risk control relative to the benchmark zero-intelligence strategies.

6 Concluding remarks

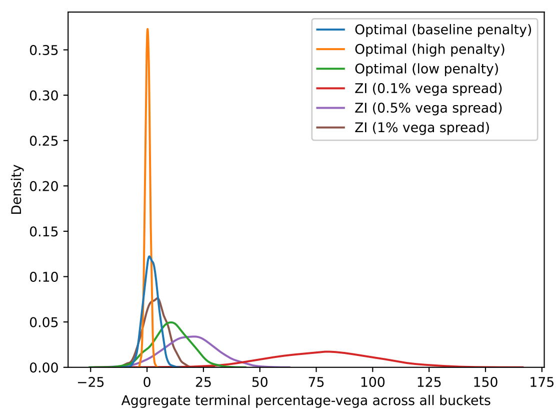
We introduce an option market making model which naturally integrates the market maker's view on the underlying volatility. Under suitable assumptions and approximations, the optimal quotes are available in closed form which can be computed efficiently. Economically, the chosen spreads are closely tied to volatility arbitrage profit originated from the actual-versus-implied volatility differential. We also demonstrate the model's inherent flexibility by showcasing how different modelling variations can be incorporated, particularly the more realistic setup where the market maker provides quotes on multiple options written on the same underlying subject to risk monitoring of customised factors. In terms of model limitations, one key assumption we made is that the



(a) Distribution of performance score.

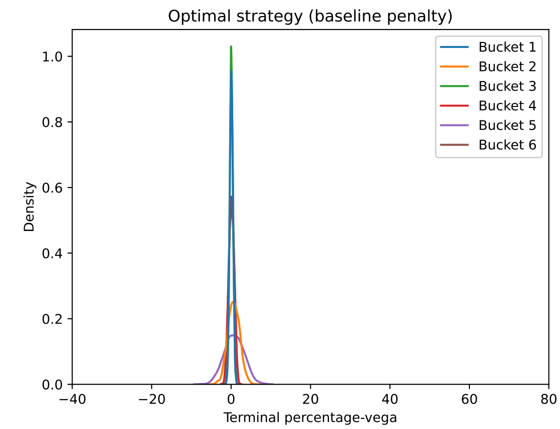


(b) Distribution of terminal portfolio value.

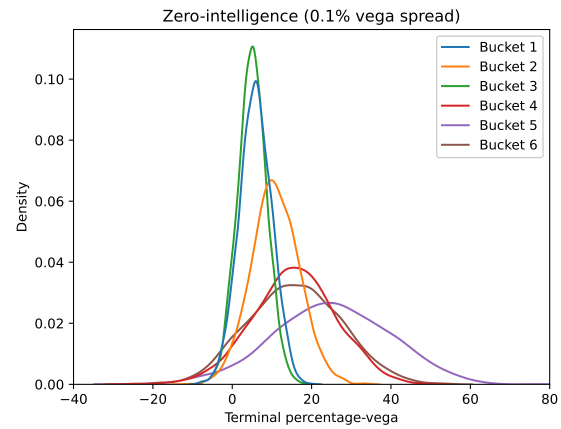


(c) Distribution of aggregate terminal percentage-vega.

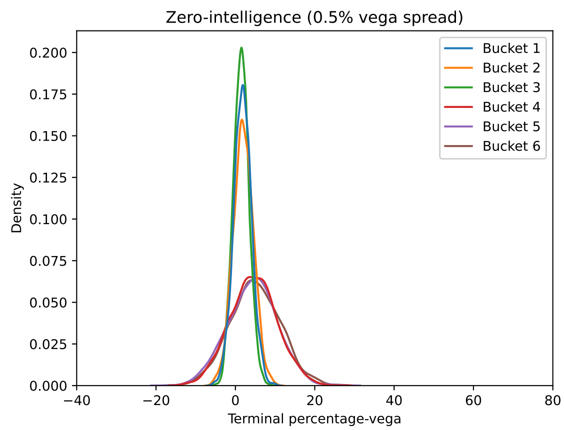
Figure 4: Empirical density functions of realised performance scores (i.e. the objective in (22)), terminal portfolio values and aggregate terminal percentage-vega across all risk buckets, under the optimal market making strategies and the zero-intelligence (ZI) strategies of quoting a constant spread around market implied volatility. In panel (b) and (c), optimal quotes under high penalty (resp. low penalty) are obtained upon scaling all the baseline values of penalty parameters α_k's by a factor of 10 (resp. 0.1).



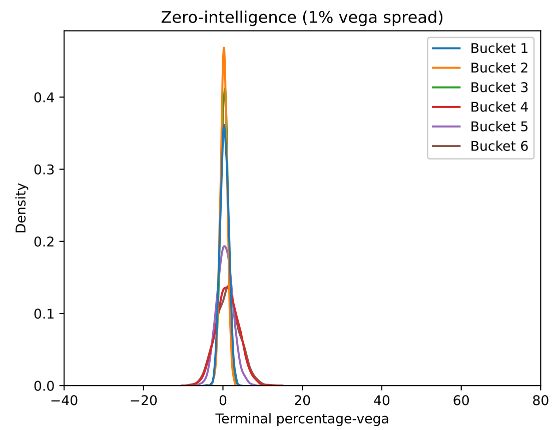
(a) Optimal market making strategies.



(b) Zero-intelligence strategy with 0.1% vega spread.



(c) Zero-intelligence strategy with 0.5% vega spread.



(d) Zero-intelligence strategy with 1% vega spread.

Figure 5: Empirical density functions of terminal percentage-vega within each risk bucket under all types of market making strategy.

market implied volatility surface is unchanged throughout the market making horizon. While this is assumption is acceptable when the market making horizon is short, consideration of more complicated implied volatility dynamics will be an interesting extension especially if one wants to extend the current framework to market making of exotic derivatives. One could also explore alternative specifications of the actual volatility process or the market order flow model that can better describe empirical data. Finally, it would be useful to investigate more rigorously the quality of the approximately-optimal quotes as well as to examine the theoretical admissibility of the proposed strategies.[12]

References

Ahmad, R. and Wilmott, P. (2005). Which free lunch would you like today, sir?: Delta hedging, volatility arbitrage and optimal portfolios. *Wilmott (November 2005)*, pages 64–79.

Avellaneda, M. and Stoikov, S. (2008). High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224.

Baldacci, B., Bergault, P., and Guéant, O. (2021). Algorithmic market making for options. *Quantitative Finance*, 21(1):85–97.

Bergault, P., Evangelista, D., Guéant, O., and Vieira, D. (2021). Closed-form approximations in multi-asset market making. *Applied Mathematical Finance*, 28(2):101–142.

Cartea, Á., Jaimungal, S., and Penalva, J. (2015). *Algorithmic and high-frequency trading*. Cambridge University Press.

Cartea, Á., Jaimungal, S., and Ricci, J. (2014). Buy low, sell high: A high frequency trading perspective. *SIAM Journal on Financial Mathematics*, 5(1):415–444.

El Aoud, S. and Abergel, F. (2015). A stochastic control approach to option market making. *Market Microstructure and Liquidity*, 1(01):1550006.

Fernandez-Tapia, J. (2015). *Modeling, optimization and estimation for the on-line control of trading algorithms in limit-order markets*. PhD thesis, Université Pierre et Marie Curie - Paris VI.

Gatheral, J. (1997). Delta hedging with uncertain volatility. In Nelken, I., editor, *Volatility in the Capital Markets: State-of-the-Art Techniques for Modeling, Managing and Trading Volatility*, chapter 5, pages 95–112. Glenlake Publishing, Fitzroy Dearborn Publishers.

Guéant, O. (2016). *The Financial Mathematics of Market Liquidity: From Optimal Execution to Market Making*, volume 33. CRC Press.

[12]As highlighted by Guéant et al. (2013), the admissibility of the approximately optimal quotes derived by Avellaneda and Stoikov (2008) is unclear. The same consideration applies to our problem with options as well.

Guéant, O. and Lehalle, C.-A. (2015). General intensity shapes in optimal liquidation. *Mathematical Finance*, 25(3):457–495.

Guéant, O., Lehalle, C.-A., and Fernandez-Tapia, J. (2013). Dealing with the inventory risk: a solution to the market making problem. *Mathematics and Financial Economics*, 7:477–507.

Ho, T. and Stoll, H. R. (1981). Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics*, 9(1):47–73.

Lai, Q., Gao, X., and Li, L. (2023). A data-driven deep learning approach for options market making. *Quantitative Finance*, 23(5):777–797.

Laruelle, S. (2013). Faisabilité de l'apprentissage des paramètres d'un algorithme de trading sur des données réelles. Available at `https://events.chairefdd.org/wp-content/uploads/2013/06/CAHIER_MICRO_1.pdf`.

Stoikov, S. and Sağlam, M. (2009). Option market making under inventory risk. *Review of Derivatives Research*, 12:55–79.

Appendix

A General intensity functions

In Section 2, we have specialised to exponential intensity function in form of $\mathbb{R} \ni \delta \mapsto \lambda^i(\delta) := \lambda_0^i e^{-\kappa^i \delta} \in \mathbb{R}$ for $i \in \{b, a\}$. The analysis can be extended to intensity functions of more general form.

For the moment, we drop the superscript $i \in \{b, a\}$ for brevity. Inspired by Guéant (2016) and Guéant and Lehalle (2015), we consider a broader class of intensity functions $\lambda(\delta)$ defined on $\delta \in \mathbb{R}$ such that: 1) $\lambda(\delta) > 0$ for all $\delta \in \mathbb{R}$; 2) $\lambda(\cdot)$ is strictly decreasing and thrice differentiable; 3) $\sup_{\delta\in\mathbb{R}} \frac{\lambda(\delta)\lambda''(\delta)}{(\lambda'(\delta))^2} < 2$; and finally 4) $\liminf_{\delta\to+\infty} \frac{\lambda(\delta)}{\lambda'(\delta)} > -\infty$.

Define $f(\delta; c) := \lambda(\delta)(\delta + c)$ and

$$H(c) = \sup_{\delta\in\mathbb{R}} f(\delta; c), \qquad c \in \mathbb{R}.$$

The first order derivative of f with respect to δ is then

$$\partial_\delta f(\delta; c) = \lambda'(\delta)\left(c + \delta + \frac{\lambda(\delta)}{\lambda'(\delta)}\right) = \lambda'(\delta)(c - g(\delta))$$

where $g(\delta) := -\delta - \frac{\lambda(\delta)}{\lambda'(\delta)}$. Under the imposed assumptions on $\delta(\cdot)$, g is twice differentiable and strictly decreasing with $g(+\infty) = -\infty$. Moreover, $c - g(-c) = \frac{\lambda(-c)}{\lambda'(-c)} < 0$. Hence by simple calculus argument one can conclude $\delta^* := g^{-1}(c) \in (-c, \infty)$ is the unique maximiser of $\sup_{\delta\in\mathbb{R}} f(\delta; c)$ such that $H(c) = f(g^{-1}(c); c)$. The first and second order derivative of H are then given by

$$H'(c) = \lambda(g^{-1}(c))\left[1 + \frac{1}{g'(g^{-1}(c))}\right] + \frac{\lambda'(g^{-1}(c))}{g'(g^{-1}(c))}(c + g^{-1}(c)),$$

$$H''(c) = \frac{1}{(g'(g^{-1}(c)))^2}\left[\lambda(g^{-1}(c))g''(g^{-1}(c)) + 2\lambda'(g^{-1}(c))(g'(g^{-1}(c)) + 1) + \lambda''(g^{-1}(c))(c + g^{-1}(c)) - H'(c)g''(g^{-1}(c))\right].$$

Revisiting the HJB equation (4), the optimal controls under generic intensity functions $\lambda^i(\cdot)$ are given by

$$\delta^{a,*} = g_a^{-1}(h(t,s,q-1) - h(t,s,q)), \qquad \delta^{b,*} = g_b^{-1}(h(t,s,q+1) - h(t,s,q)),$$

and the HJB equation becomes

$$\begin{aligned}\partial_t h + \frac{\sigma^2(t,s) - \sigma_{imp}^2}{2}\Gamma^{\$}(t,s)q - \beta q^2 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}h + H_a(h(t,s,q-1) - h(t,s,q))& \\ + H_b(h(t,s,q+1) - h(t,s,q)) = 0.&\end{aligned}$$

In the above, $g_i(\cdot)$ and $H_i(\cdot)$ for $i \in \{b,a\}$ are defined similarly as before based on a given intensity function $\lambda^i(\delta)$. The second-order approximation now yields

$$\begin{aligned}\partial_t h + \frac{\sigma^2(t,s) - \sigma_{imp}^2}{2}\Gamma^{\$}(t,s)q - \beta q^2 + \frac{\sigma^2(t,s)}{2}s^2\partial_{ss}h& \\ + H_a(0) + H_a'(0)(h(t,s,q-1) - h(t,s,q)) + \frac{H_a''(0)}{2}(h(t,s,q-1) - h(t,s,q))^2& \\ + H_b(0) + H_b'(0)(h(t,s,q+1) - h(t,s,q)) + \frac{H_b''(0)}{2}(h(t,s,q+1) - h(t,s,q))^2 = 0.&\end{aligned}$$

The same ansatz $h(t,s,q) = \psi_0(t,s) + \psi(t,s)q + \psi_2(t)q^2$ can be used to solve the above equation.